Business Loves Uplift

Causal Impact Evaluation With Python

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ODSC AI West

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decision academy

Agenda

- Causal Identification
- A/B Tests
- Clustered A/B Tests
- Double Machine Learning
- Synthetic Control
- Triangulation

Causal Identification:

What is uplift?

Causal Identification

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Causal Identification

- Imagine we're interested in **balloons**

Causal Identification

- Imagine we're interested in **balloons** and want to **test a new burner**

Causal Identification

- Imagine we're interested in balloons and want to test a new burner
- We have **two balloons - observer balloon** where we sit

observer balloon (us)

Causal Identification

- Imagine we're interested in balloons and want to test a new burner
- We have **two balloons - observer balloon** where we sit and a **test balloon** with a new burner

observer balloon (us)

test balloon (new burner)

Causal Identification

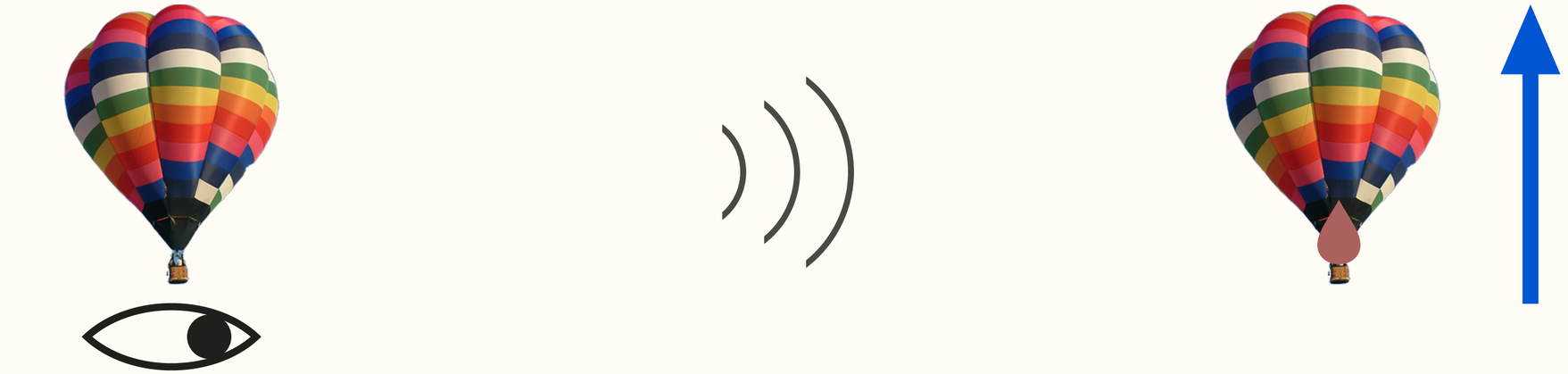
- Imagine we're interested in balloons and want to test a new burner
- We have two balloons - observer balloon where we sit and a test balloon with a new burner
- We communicate with the **test balloon** crew over the radio, ask them to **fire the burner**

observer balloon (us)

test balloon (new burner)

Causal Identification

- Imagine we're interested in balloons and want to test a new burner
- We have two balloons - observer balloon where we sit and a test balloon with a new burner
- We communicate with the **test balloon** crew over the radio, ask them to **fire the burner**, and observe **how fast the test balloon moves up**



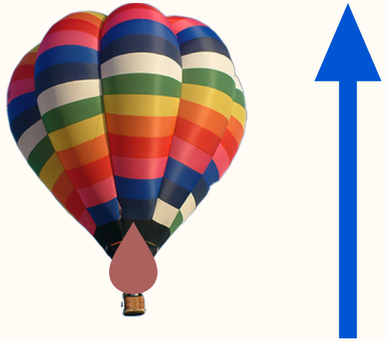
observer balloon (us)

test balloon (new burner)

Causal Identification

- **What do we learn** from this observation about the new burner?

observer balloon (us)

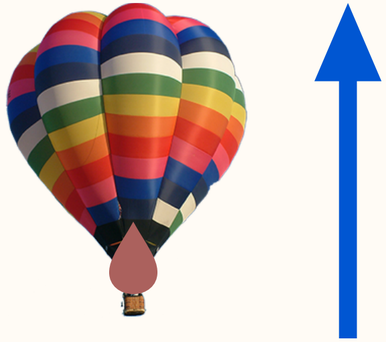


test balloon (new burner)

Causal Identification

- What do we learn from this observation about the new burner?
- It seems we **don't learn much**. We need some sort of comparison to say whether the new burner is better than the old one

observer balloon (us)



test balloon (new burner)

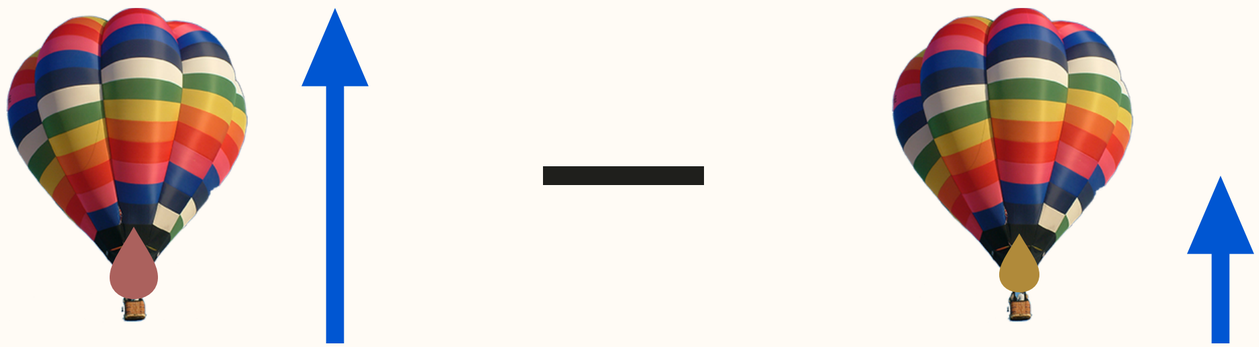
Causal Identification

- What we call **uplift** is the **difference** between the **speed** of our **test balloon** moving up on a **new burner**

new burner

Causal Identification

- What we call **uplift** is the **difference** between the **speed** of our **test balloon** moving up on a **new burner** vs the **speed** of the same **test balloon** in the same exact circumstances on the **old burner**



new burner

old burner

Causal Identification

- What we call uplift is the difference between the speed of our test balloon moving up on a new burner vs the speed of the same test balloon in the same exact circumstances on the old burner

HOUSTON, WE'VE GOT A PROBLEM!

new burner

old burner

Causal Identification

- Mathematically, we often express uplift as **$Y_i(1) - Y_i(0)$**, where i is our test balloon, Y is how fast it moves up, 1 stands for the new burner and 0 for the old one

$$ITE = Y_i(1) - Y_i(0)$$

$Y_i(1)$

$Y_i(0)$

Causal Identification

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Potential outcomes

$Y_i(1)$

$Y_i(0)$

Causal Identification

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Causal Identification

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$$ITE = Y_i(1) - Y_i(0)$$

Potential outcomes

In reality we can only **observe one** of them at the same time, the other remains a **counterfactual**

$Y_i(1)$

$Y_i(0)$

Causal Identification

- Mathematically, we often express uplift as $\mathbf{Y_i(1) - Y_i(0)}$, where i is our test balloon, Y is how fast it moves up, 1 stands for the new burner and 0 for the old one

other remains a **counterfactual**

$Y_i(1)$

$Y_i(0)$

Causal Identification

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SOLUTION?

Causal Identification

- The **intuitive response** to the fundamental problem is to **compare** our test balloon **against something**

Causal Identification

- The intuitive response to the fundamental problem is to compare our test balloon against something
- **Intuition 1**: **compare** the test balloon **to itself** with the old burner (a **pre-post comparison**)

Intuition 1: pre-post

Causal Identification

- The intuitive response to the fundamental problem is to compare our test balloon against something
- Intuition 1: compare the test balloon to itself with the old burner (a pre-post comparison)
- **Intuition 2**: **compare** the test balloon **to another balloon** with an old burner (an **external control comparison**)

Intuition 1: pre-post

Intuition 2: external control

Causal Identification

Intuition 1

- In the pre-post comparison we look at the **same object** at **two** (or more) **different occasions**

yesterday, old burner

today, new burner

Causal Identification

Intuition 1

- In the pre-post comparison we look at the same object at two different occasions
- **Problem?**

yesterday, old burner

today, new burner

Causal Identification

Intuition 1

- In the pre-post comparison we look at the same object at two different occasions
- Problem?
- **Many things can differ between these two occasions** beside the change whose effects we want to observe

yesterday, old burner

today, new burner

Causal Identification

Intuition 1

- In the pre-post comparison we look at the same object at two different occasions
- Problem?
- Many things can differ between these two occasions beside the change whose effects we want to observe
- These **accidental factors** can make it **impossible to isolate** the **effect** of our **treatment** from the effects of other changes

yesterday, old burner

today, new burner

Causal Identification

Intuition 1: Example

- We ran a **new marketing campaign** in **May**

Causal Identification

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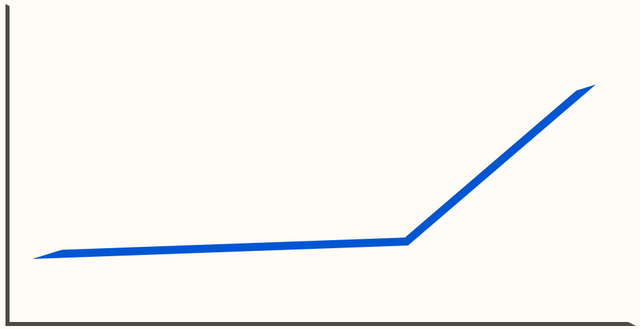
Intuition 1: Example

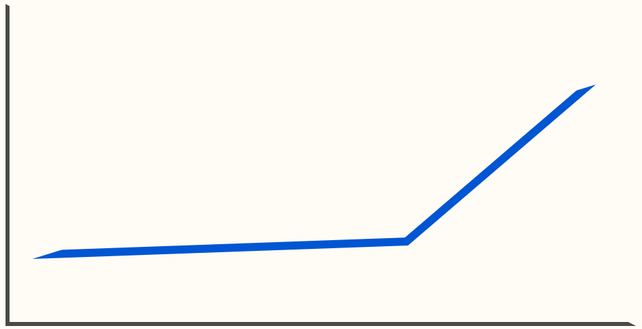
- We ran a new marketing campaign in May
- **Sales** went **up** in **May** compared to **April** (and/or May last year)



Causal Identification

Intuition 1: Example

- We ran a new marketing campaign in May
- Sales went up in May compared to April and May last year
- **Did the campaign work?**



Causal Identification

Intuition 1: Example

- We ran a new marketing campaign in May
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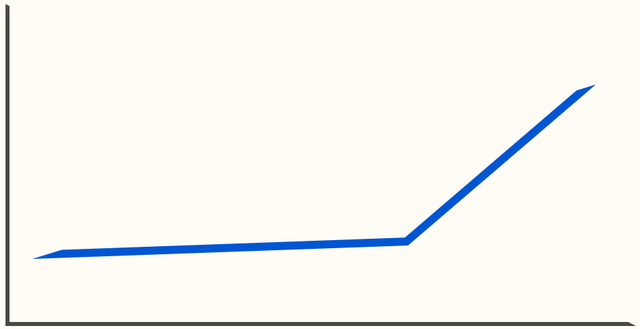
- Did the campaign work?
- **We don't know**, possibly:



Causal Identification

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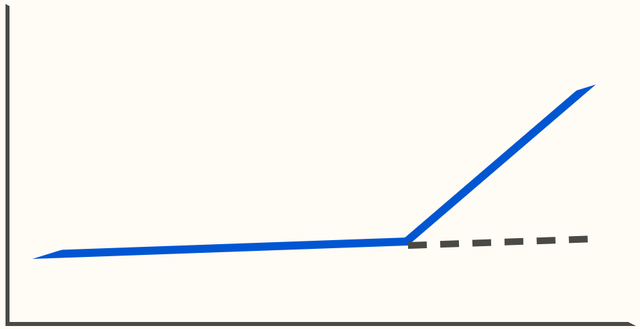

It worked, sales would not move otherwise

Solid: sales we observed. Dashed: sales without the campaign

Causal Identification

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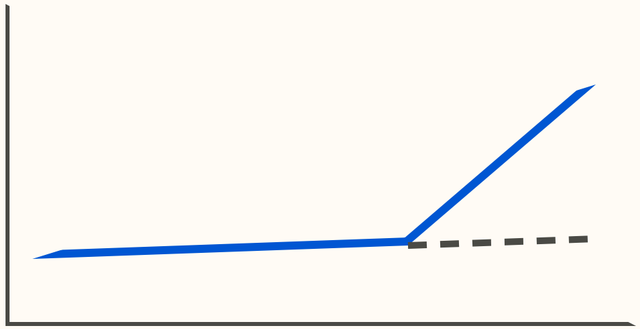

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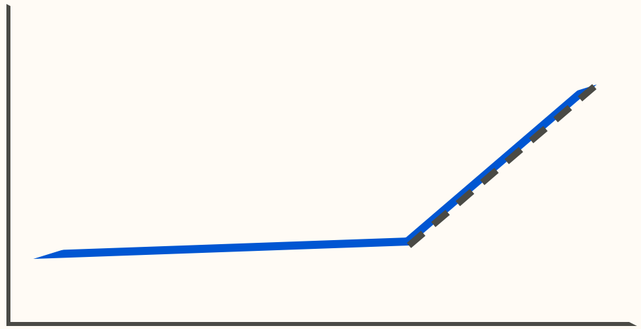

It did not work, all movement came from the market movement

Solid: sales we observed. Dashed: sales without the campaign

Causal Identification

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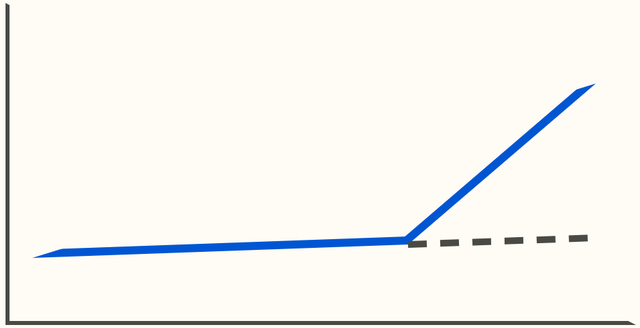

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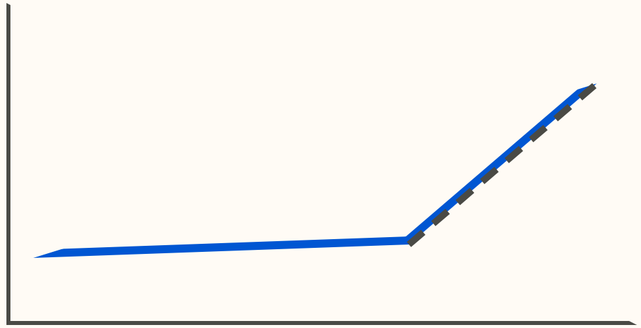

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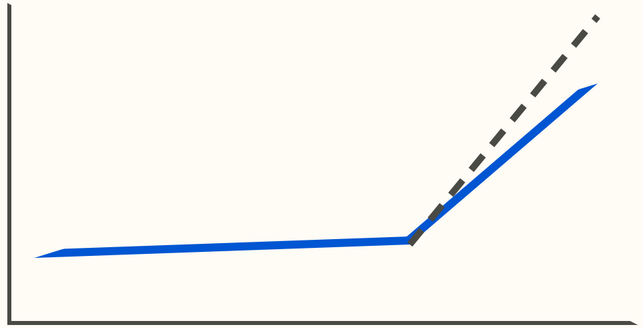


It was harmful - the movement we observed came from the market movement, but the campaign slowed it down; we'd have even more sales without the campaign

Solid: sales we observed. Dashed: sales without the campaign

Causal Identification

Intuition 2

- In the **external comparison** we look at two **different objects** at **the same occasion**

test balloon

another balloon

Causal Identification

Intuition 2

- In the external comparison we look at two different objects at the same occasion
- **Problem?**

test balloon

another balloon

Causal Identification

Intuition 2

- In the external comparison we look at two different objects at the same occasion
- Problem?
- **Many things can differ between these two objects** beside the change whose effects we want to observe

test balloon

another balloon

Causal Identification

Intuition 2

- In the external comparison we look at two different objects at the same occasion
- Problem?
- Many things can differ between these two objects beside the change whose effects we want to observe
- These **differences** can make it **impossible to isolate** the effect of our **treatment** from the effects of everything else that makes the two objects different

test balloon

another balloon

Causal Identification

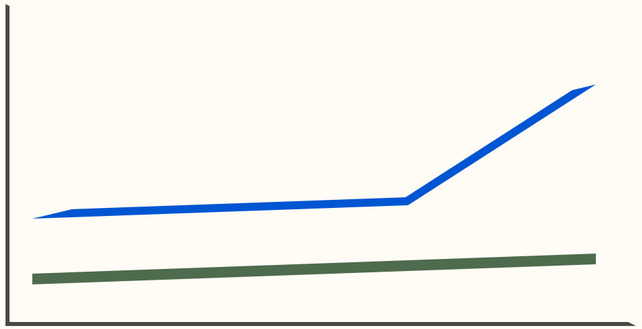
Intuition 2: Example

- We ran a **new marketing campaign in May** in the **North** region, but **not** in the **South**

Causal Identification

Intuition 2: Example

- We ran a new marketing campaign in May in the North region, but not in the South
- **Sales** went **up** more in the **North** than in the South



Blue: North (campaign). Green: South (no campaign).

Causal Identification

Intuition 2: Example

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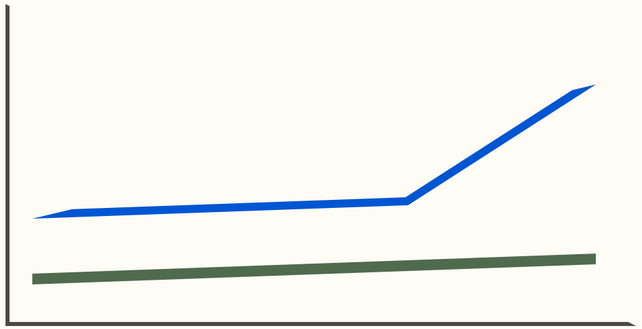

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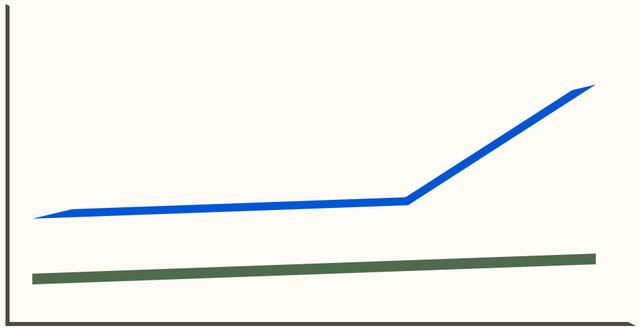

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Causal Identification

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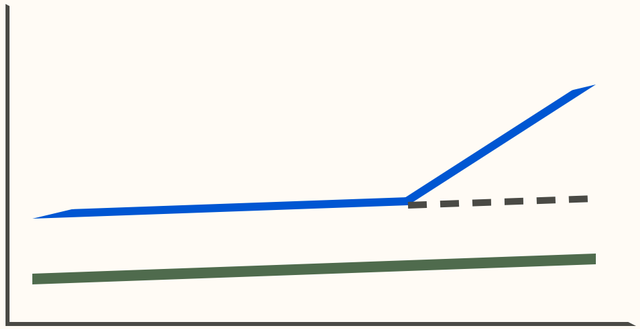

It worked, sales in the North would not move more than in the South otherwise

Blue: North (campaign). Green: South (no campaign). Dashed: North without the campaign

Causal Identification

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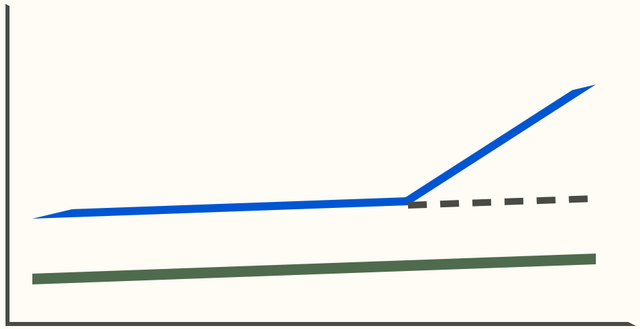

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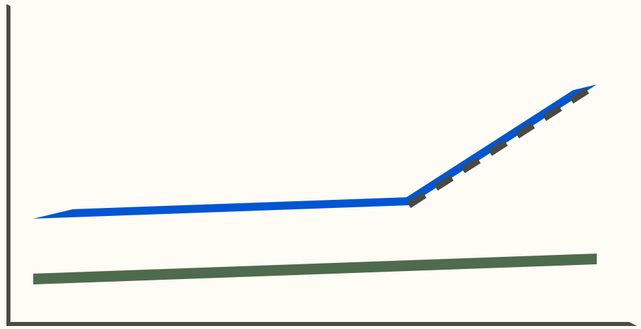

It did not work, the North was growing faster anyway (perhaps even that's why we picked it for the campaign in the first place)

Blue: North (campaign). Green: South (no campaign). Dashed: North without the campaign

Causal Identification

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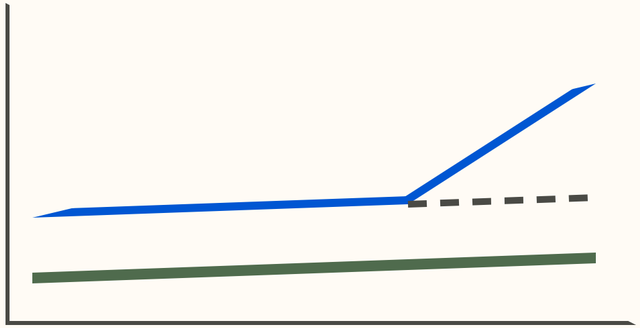

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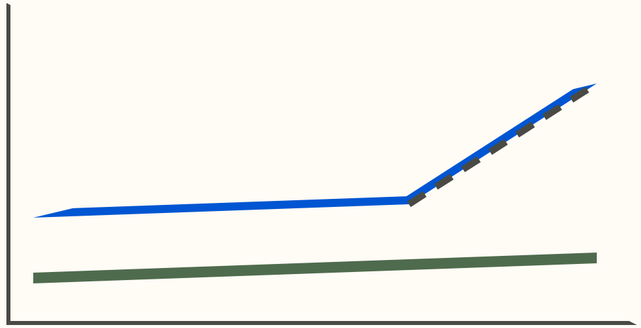

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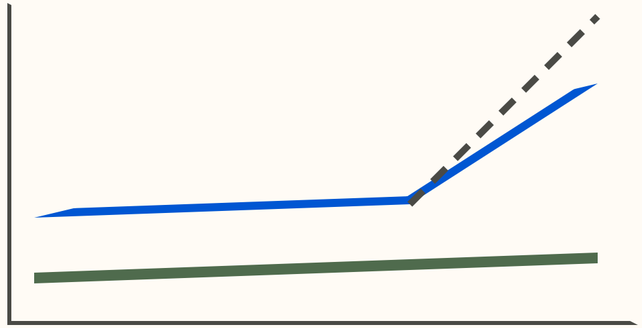

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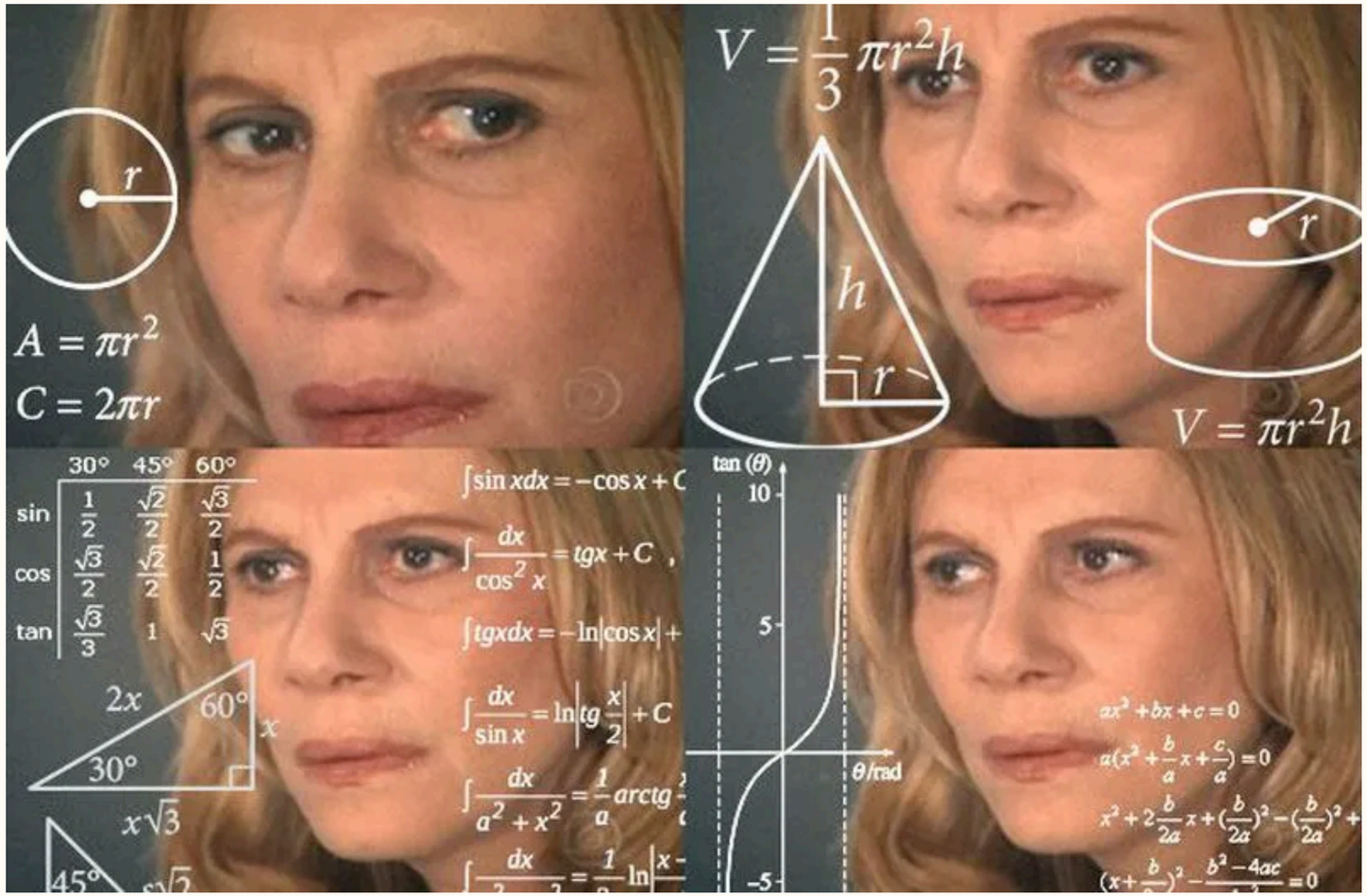
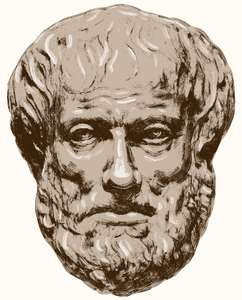
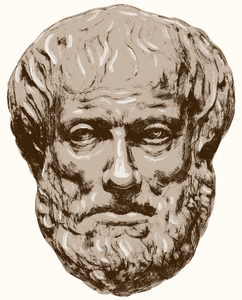

$A = \pi r^2$
$C = 2\pi r$
$V = \frac{1}{3}\pi r^2 h$
$V = \pi r^2 h$
30° 45° 60°
sin $\frac{1}{2}$ $\frac{\sqrt{2}}{2}$ $\frac{\sqrt{3}}{2}$
cos $\frac{\sqrt{3}}{2}$ $\frac{\sqrt{2}}{2}$ $\frac{1}{2}$
tan $\frac{\sqrt{3}}{3}$ 1 $\sqrt{3}$
$\int \sin x dx = -\cos x + C$
$\int \frac{dx}{\cos^2 x} = tgx + C$
$\int tgx dx = -\ln|\cos x| +$
$\int \frac{dx}{\sin x} = \ln\left|tg\frac{x}{2}\right| + C$
$\int \frac{dx}{a^2 + x^2} = \frac{1}{a} arctg$
tan (θ)
θ/rad
$ax^2 + bx + c = 0$
$a(x^2 + \frac{b}{a}x + \frac{c}{a}) = 0$

NO WAY OUT?

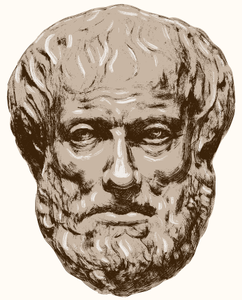


RCT /
A/B Test



RCT /
A/B Test

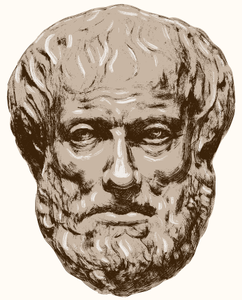
Quasi-
experiments

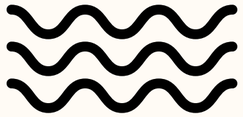


RCT /
A/B Test

Quasi-
experiments

Adjustment-
based
methods





RCT /
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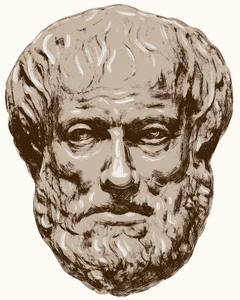

A/B Testing

A/B Testing

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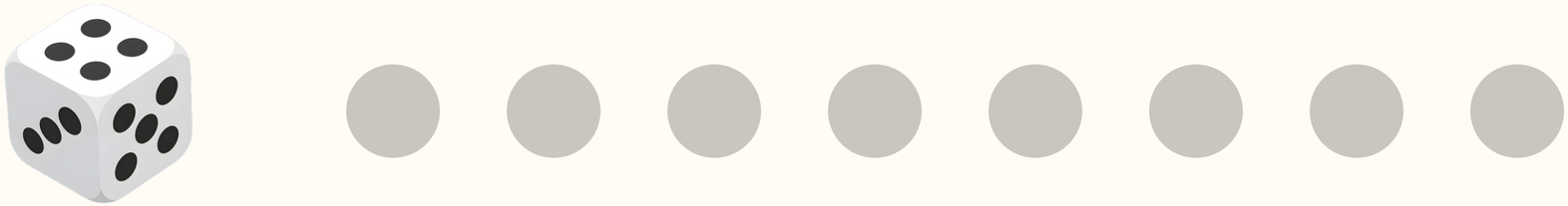
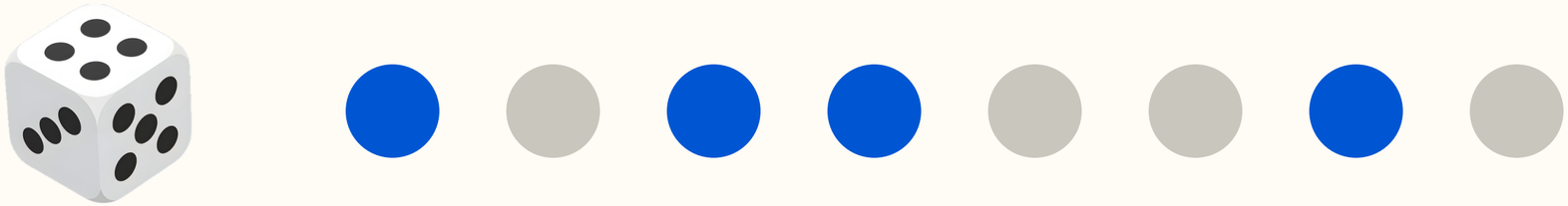
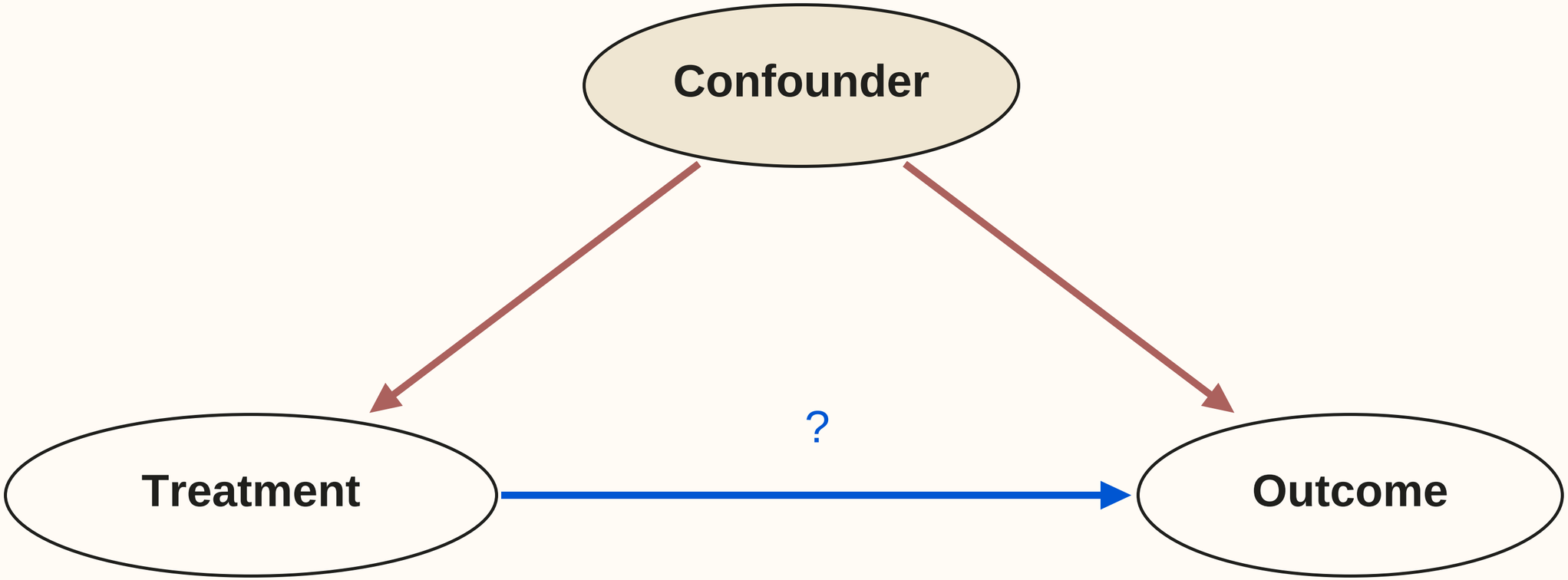


A/B Testing



blue: group A (treatment) grey: group B (control)

A/B Testing



A/B Testing

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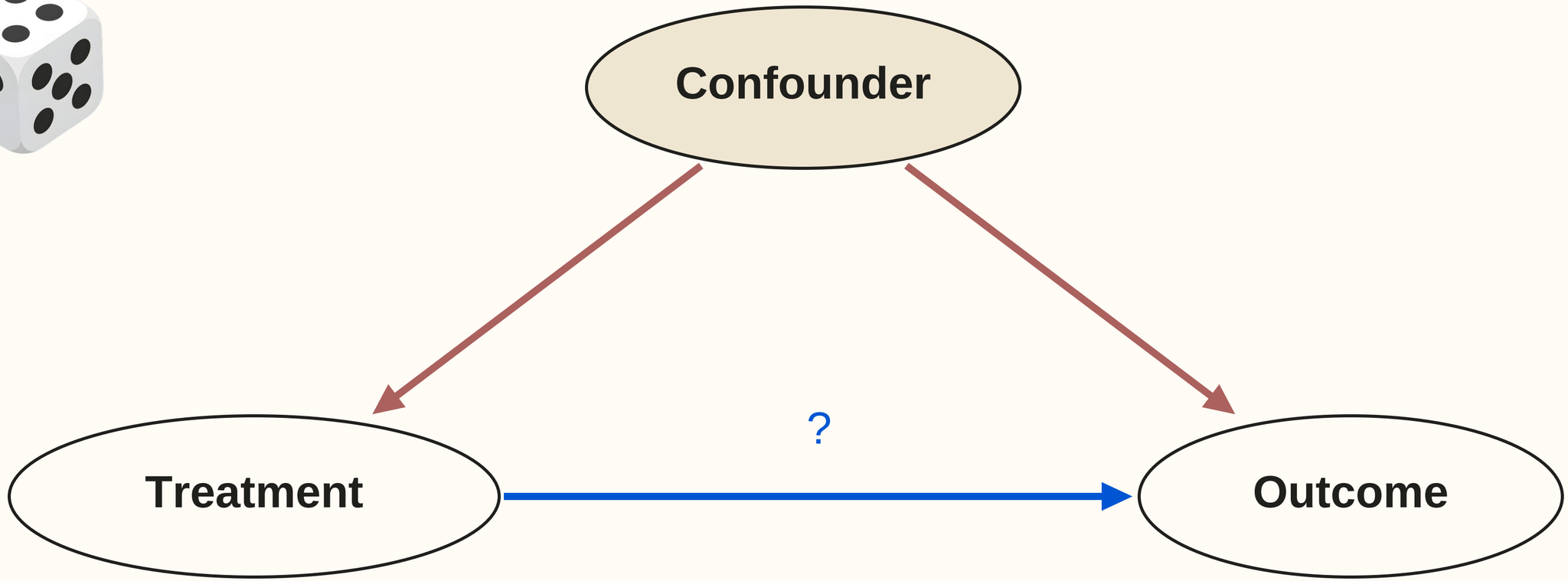
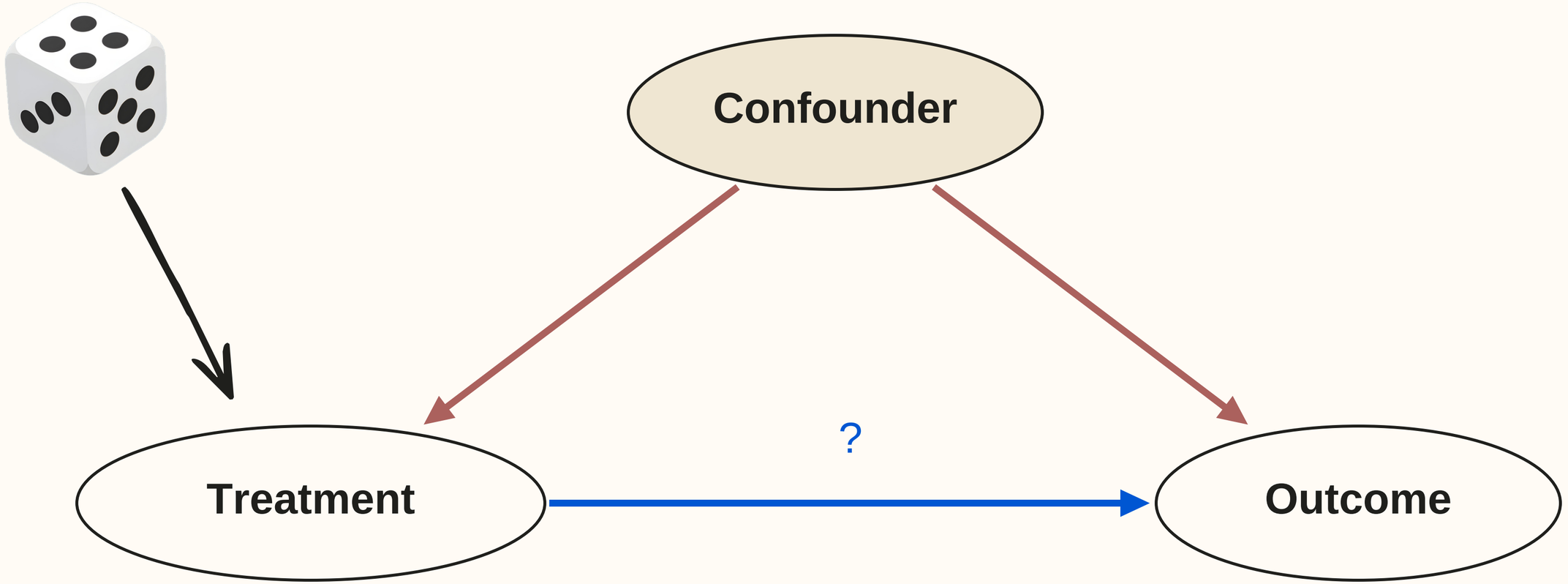
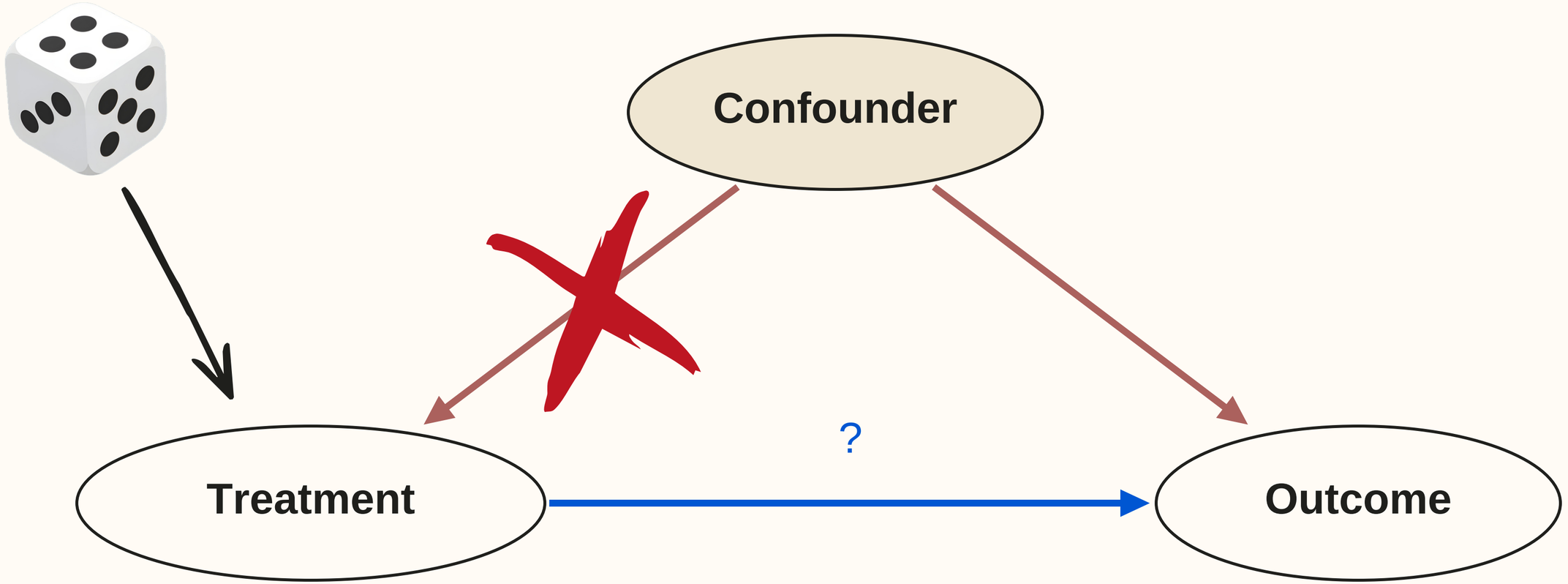


A/B Testing



A/B Testing



A/B Testing

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$$ATE = \frac{1}{n} \sum_{i}^{n} \left(Y_i \left(1 \right) - Y_i \left(0 \right) \right)$$

A/B Testing

decision academy

$$ATE = \frac{1}{n}\sum_{i}^{n}\left(Y_i\left(1\right) - Y_i\left(0\right)\right)$$

$$ATE = E\left[P\left(Y\left|do\left(T=1\right)\right) - P(Y|do\left(T=0\right)\right)\right]$$

A/B Testing

decision academy

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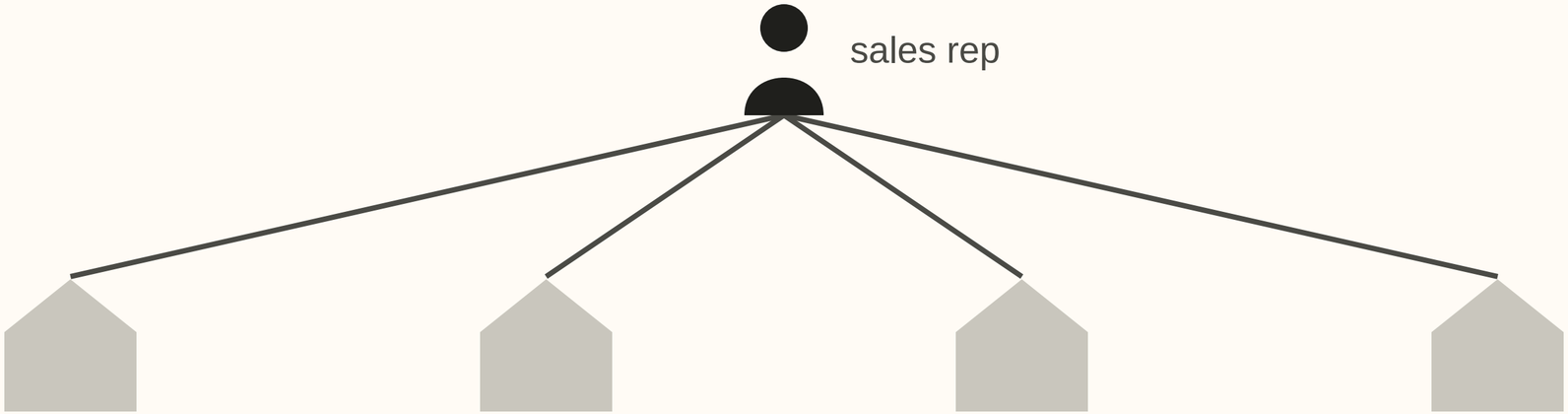
do-operator

decision academy

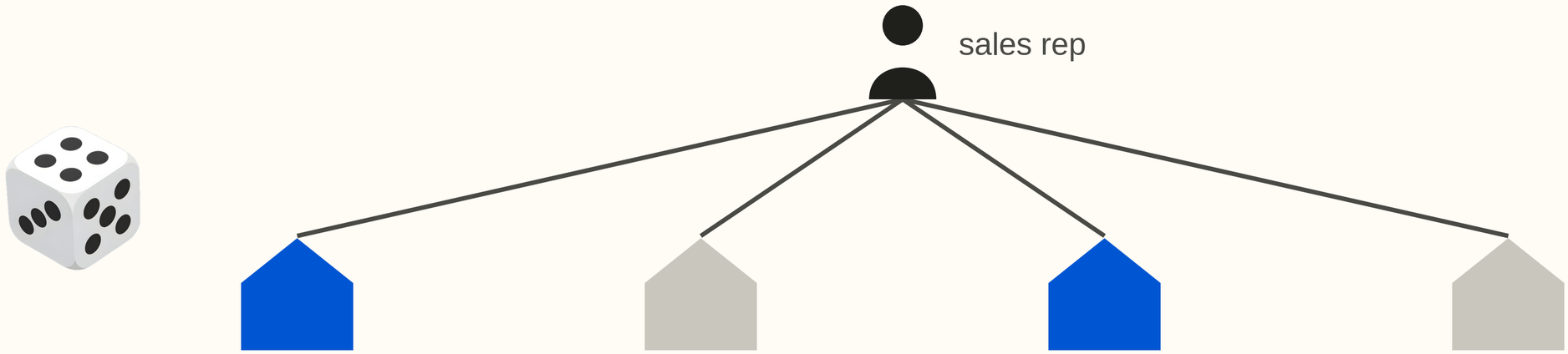
Let’s design an A/B test

Clustered A/B Test

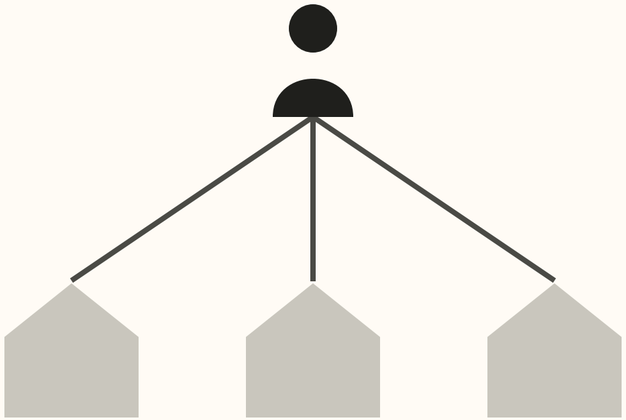
Clustered A/B Test

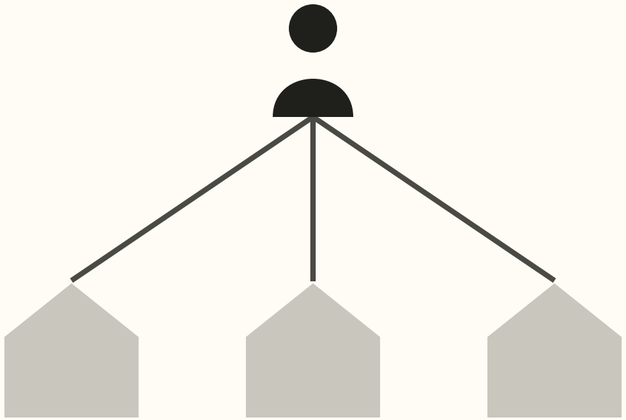


Clustered A/B Test

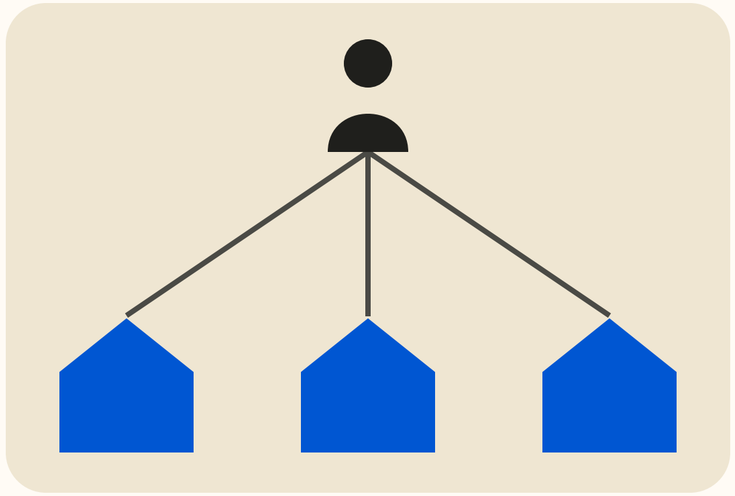


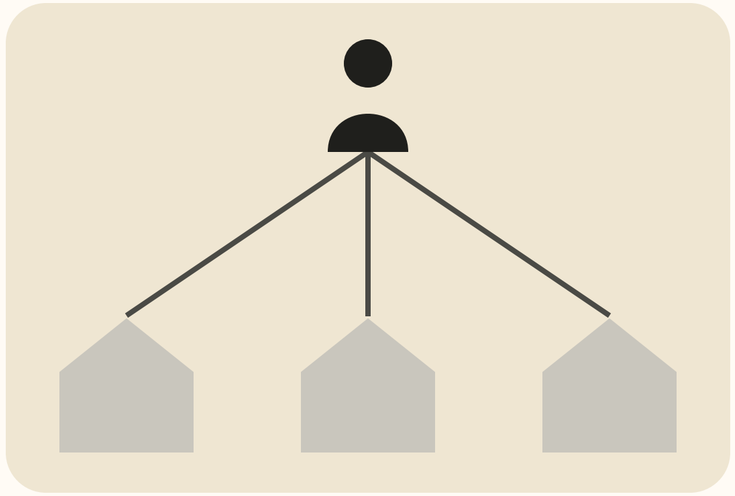
Clustered A/B Test





Clustered A/B Test





Clustered A/B Test

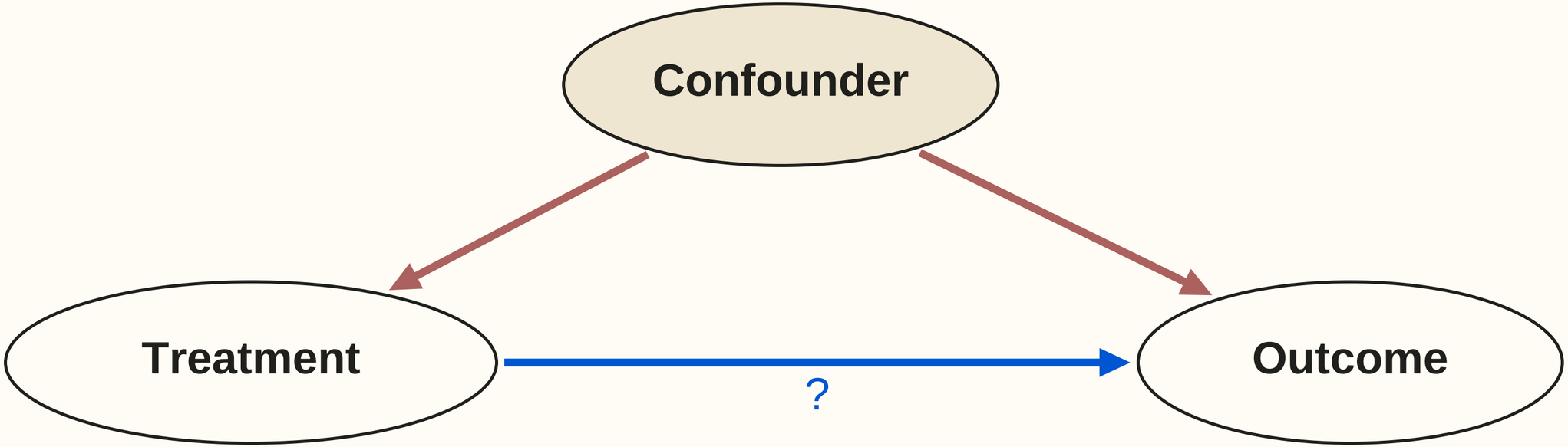
decision academy

Let’s design a clustered A/B test

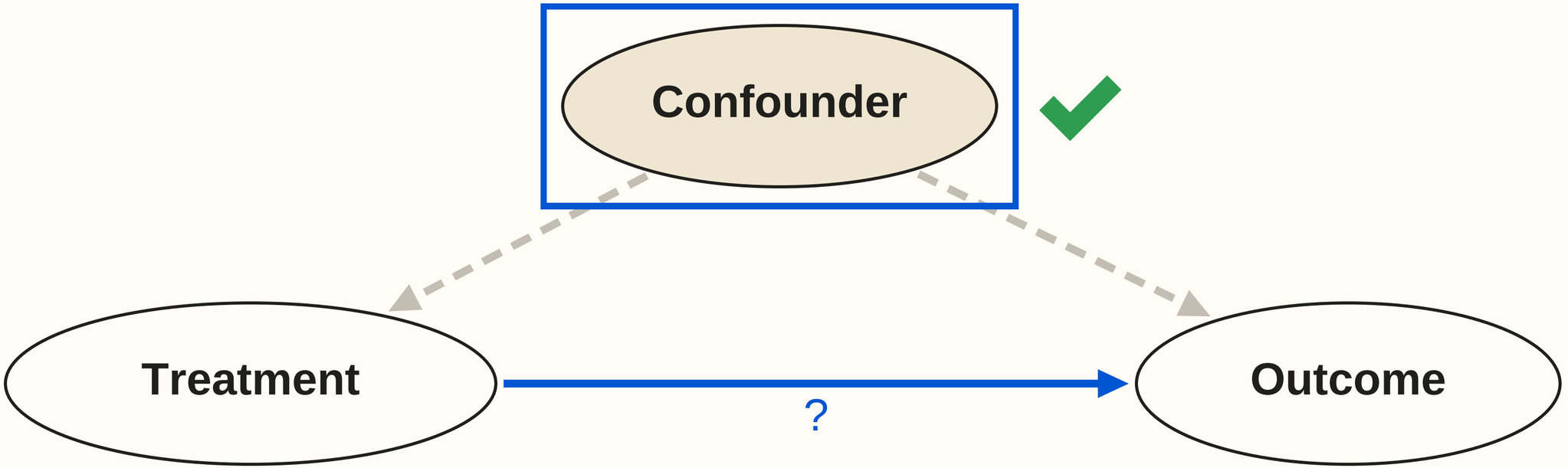
Clustered A/B Test

Double Machine Learning

Double Machine Learning



Double Machine Learning



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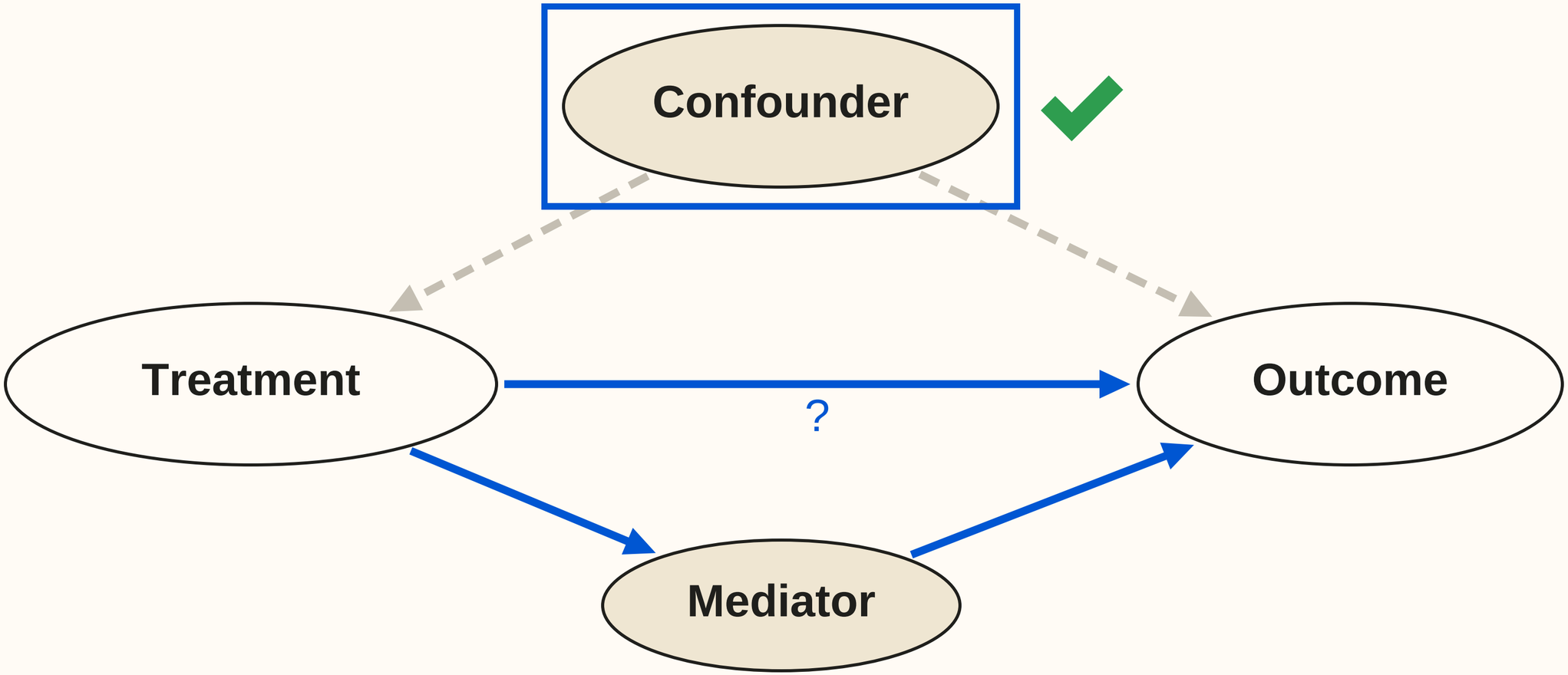
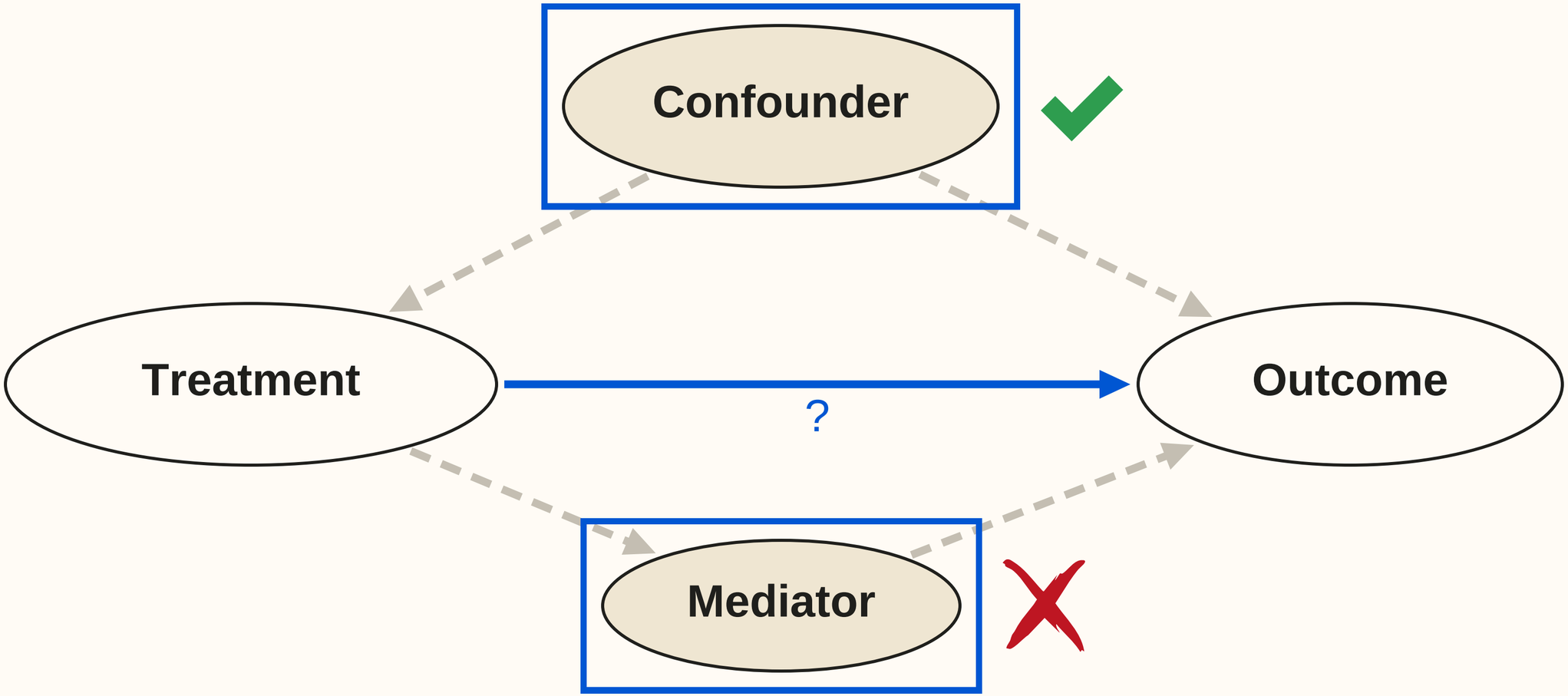
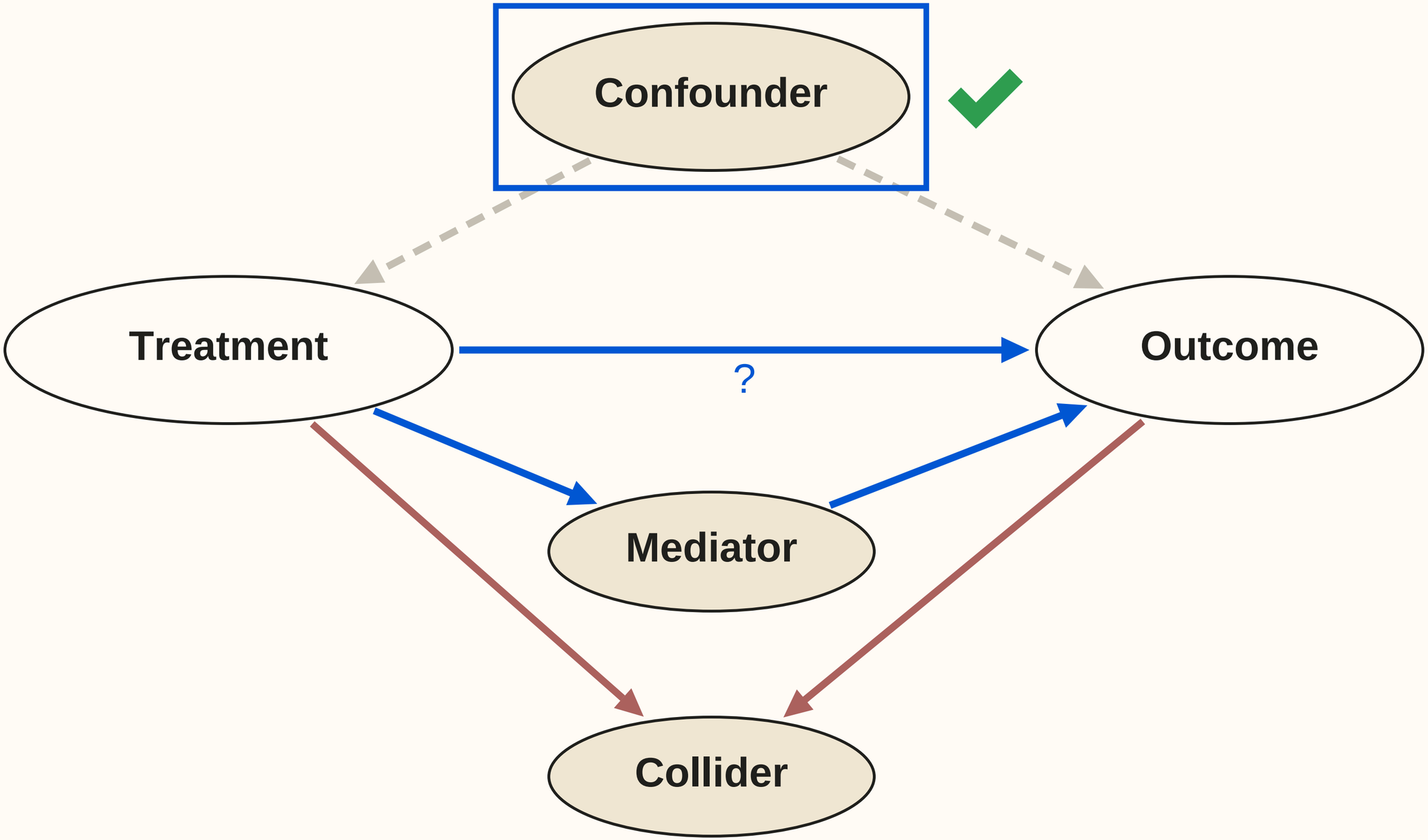


Double Machine Learning

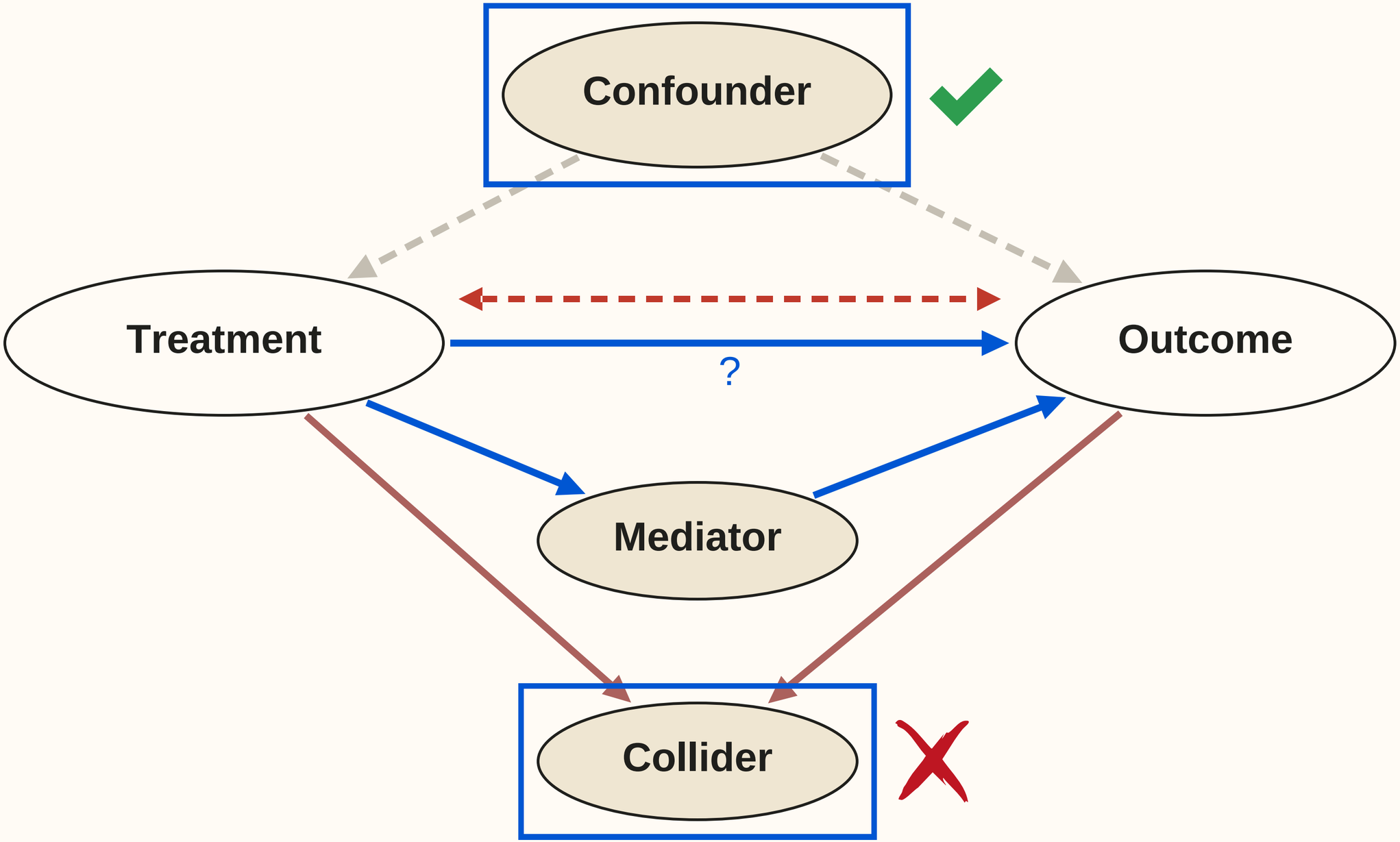


Double Machine Learning

decision academy



Double Machine Learning

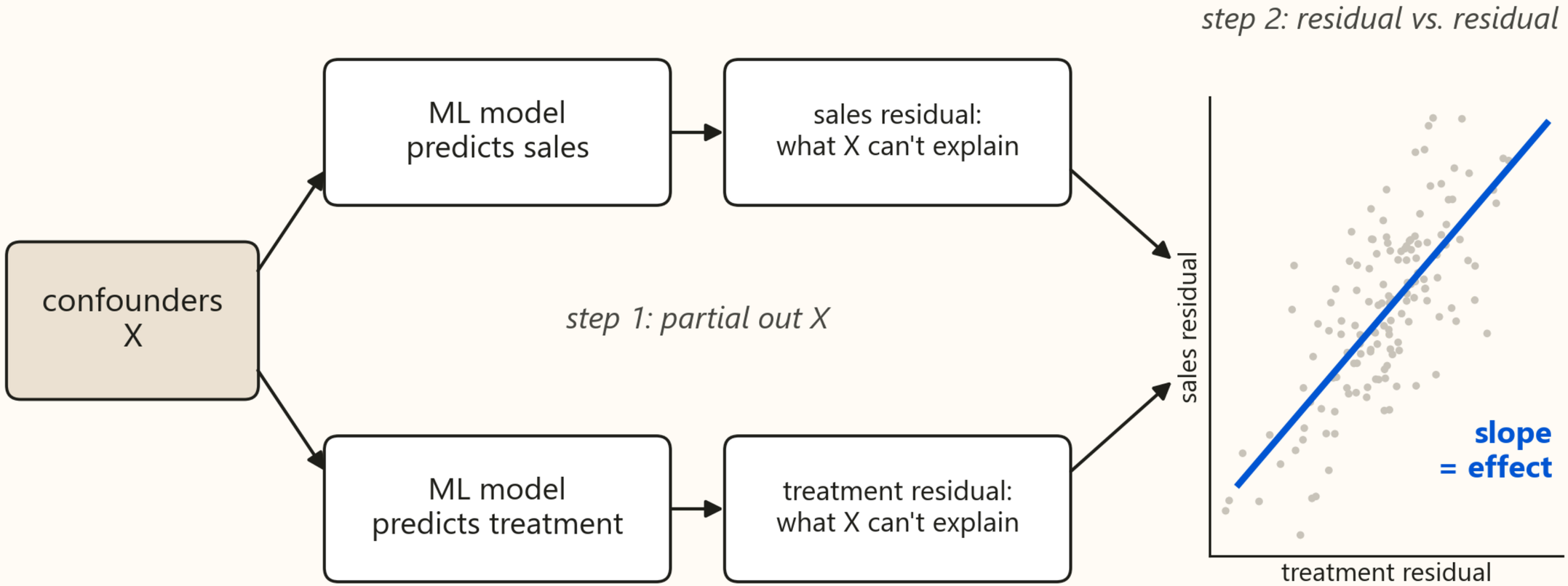


decision academy

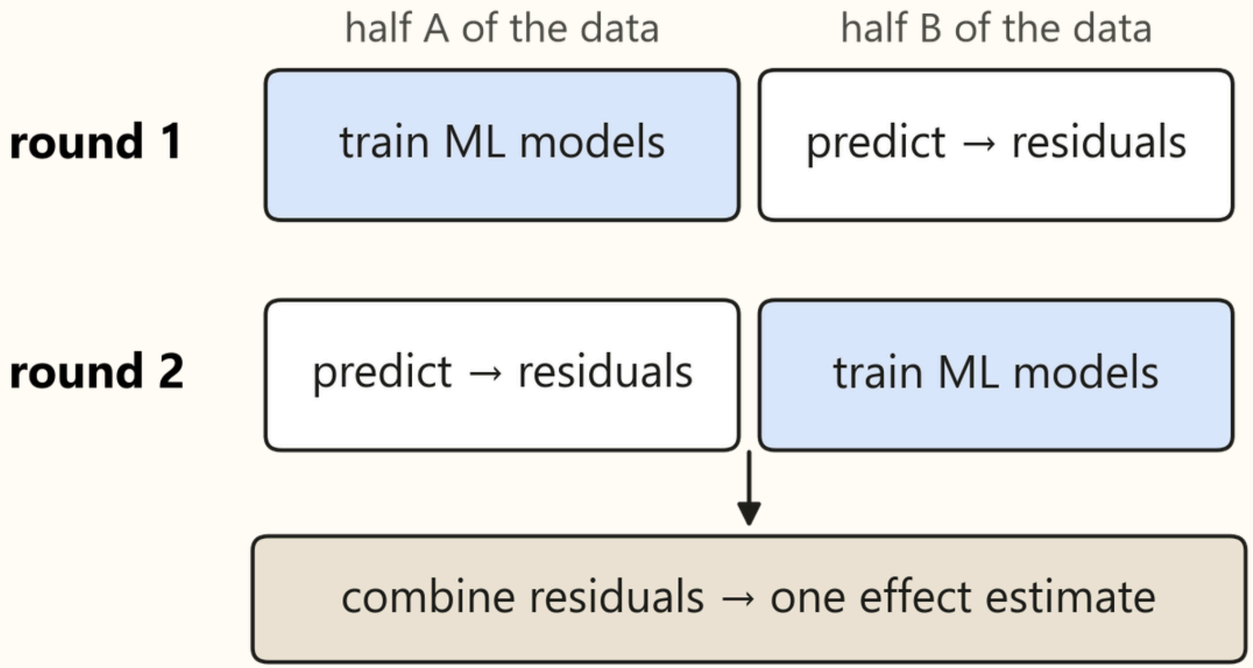
More Features ≠ Better Identification

- Identification comes from **the graph** rather than the number of features
- DML handles **estimation,** not identification. It can't fix a wrong adjustment set

Double Machine Learning



Double Machine Learning



no observation is predicted by a model that saw it

Double Machine Learning

decision academy

Estimand: CATE

- With DML we can estimate the **CATE**: the average effect for units with features X = x
- **Average the CATE** over everyone and you get the **ATE**
- That's what lets us compare DML with the A/B test and SC

$$\mathrm{CATE}(x) = \mathbb{E}\left[Y(1) - Y(0) \mid X = x\right]$$

$$\mathrm{CATE}(x) = \mathbb{E}\left[Y \mid do(T=1),\ X = x\right] - \mathbb{E}\left[Y \mid do(T=0),\ X = x\right]$$

$$\mathrm{ATE} = \mathbb{E}_X\left[\mathrm{CATE}(X)\right]$$

$$Y = \theta(X) \cdot T + g(X) + \varepsilon$$

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More Features ≠ Better Identification

- Identification comes from **the graph** rather than the number of features
- DML handles **estimation,** not identification. It can't fix a wrong adjustment set

WE CONTROLLED FOR ALL CONFOUNDERS

WE CONTROLLED FOR
ALL CONFOUNDERS

Double Machine Learning

Sensitivity: partial R^2

Double Machine Learning

decision academy

Sensitivity: partial R^2

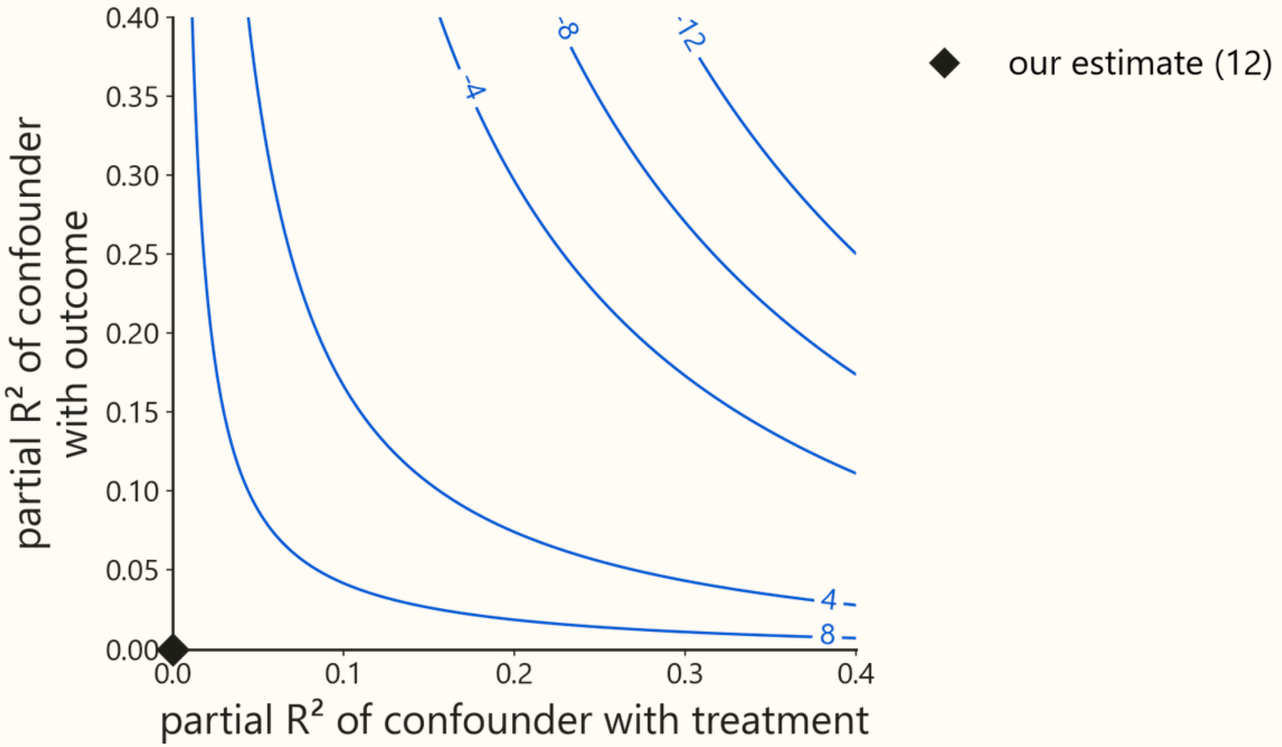
- What if there's a **confounder we didn't measure**?

Double Machine Learning

decision academy

Sensitivity: partial R^2

- What if there's a confounder we didn't measure?
- Each point is a hidden confounder: how much it explains of **treatment (x)** and **outcome (y)**



Double Machine Learning

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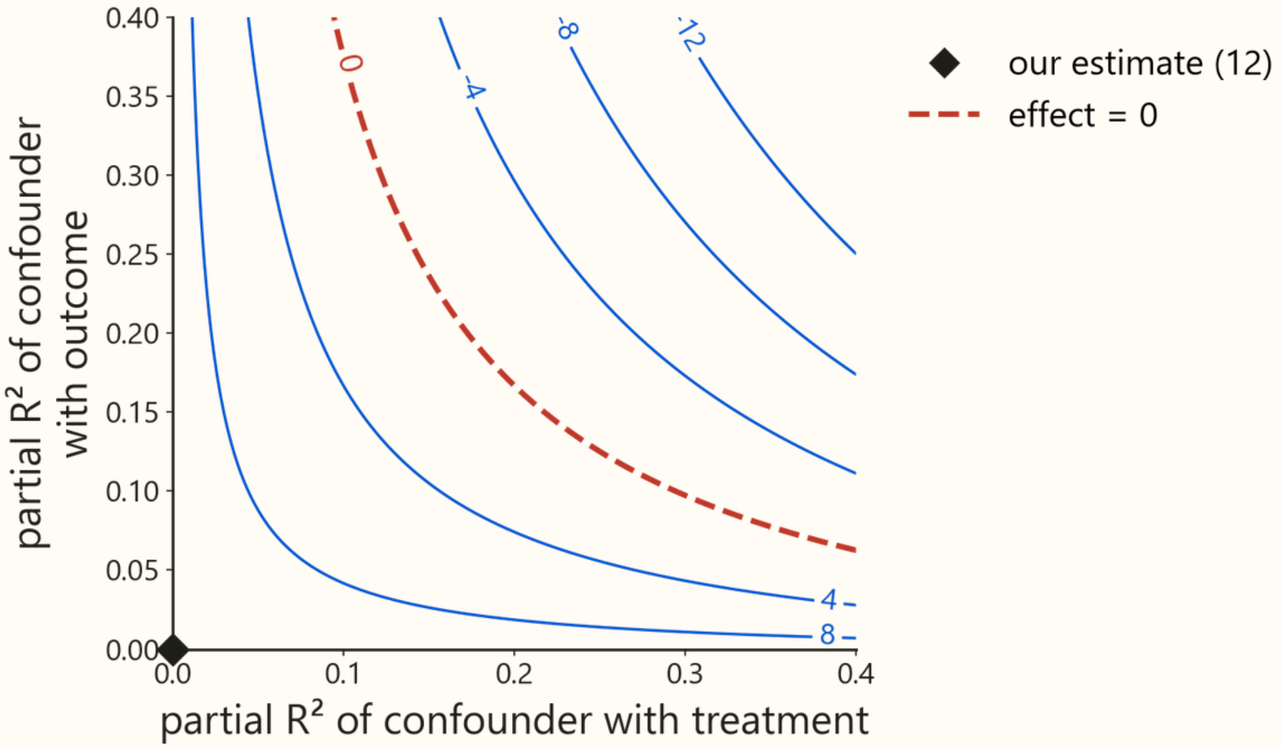
Sensitivity: partial R^2

- What if there's a confounder we didn't measure?
- Each point is a hidden confounder: how much it explains of treatment (x) and outcome (y)
- Beyond the **red line**, such a confounder would wipe out the effect



Double Machine Learning

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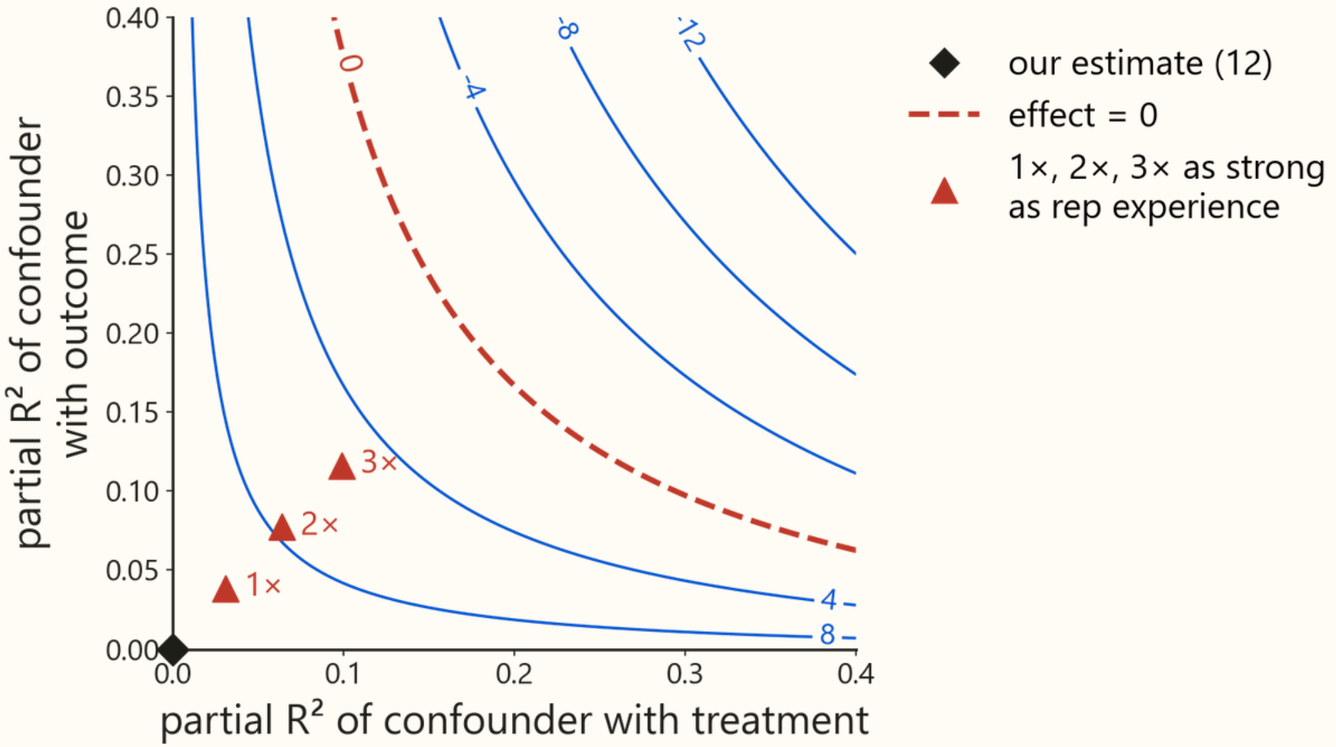
Sensitivity: partial R^2

- What if there's a confounder we didn't measure?
- Each point is a hidden confounder: how much it explains of treatment (x) and outcome (y)
- Beyond the red line, such a confounder would wipe out the effect
- Triangles: **1×, 2×, 3× as strong** as rep experience, still below the line



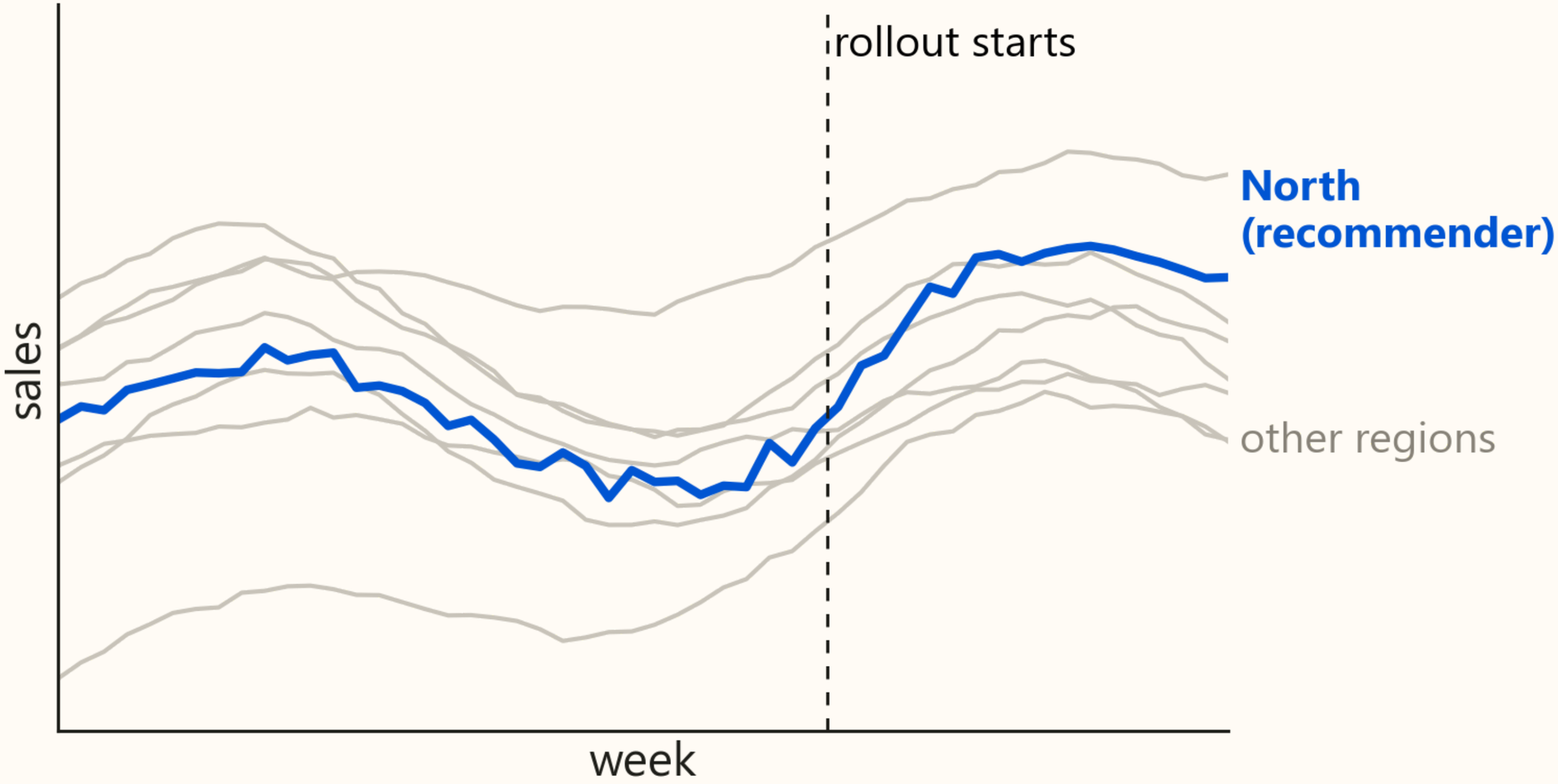
Double Machine Learning

decision academy

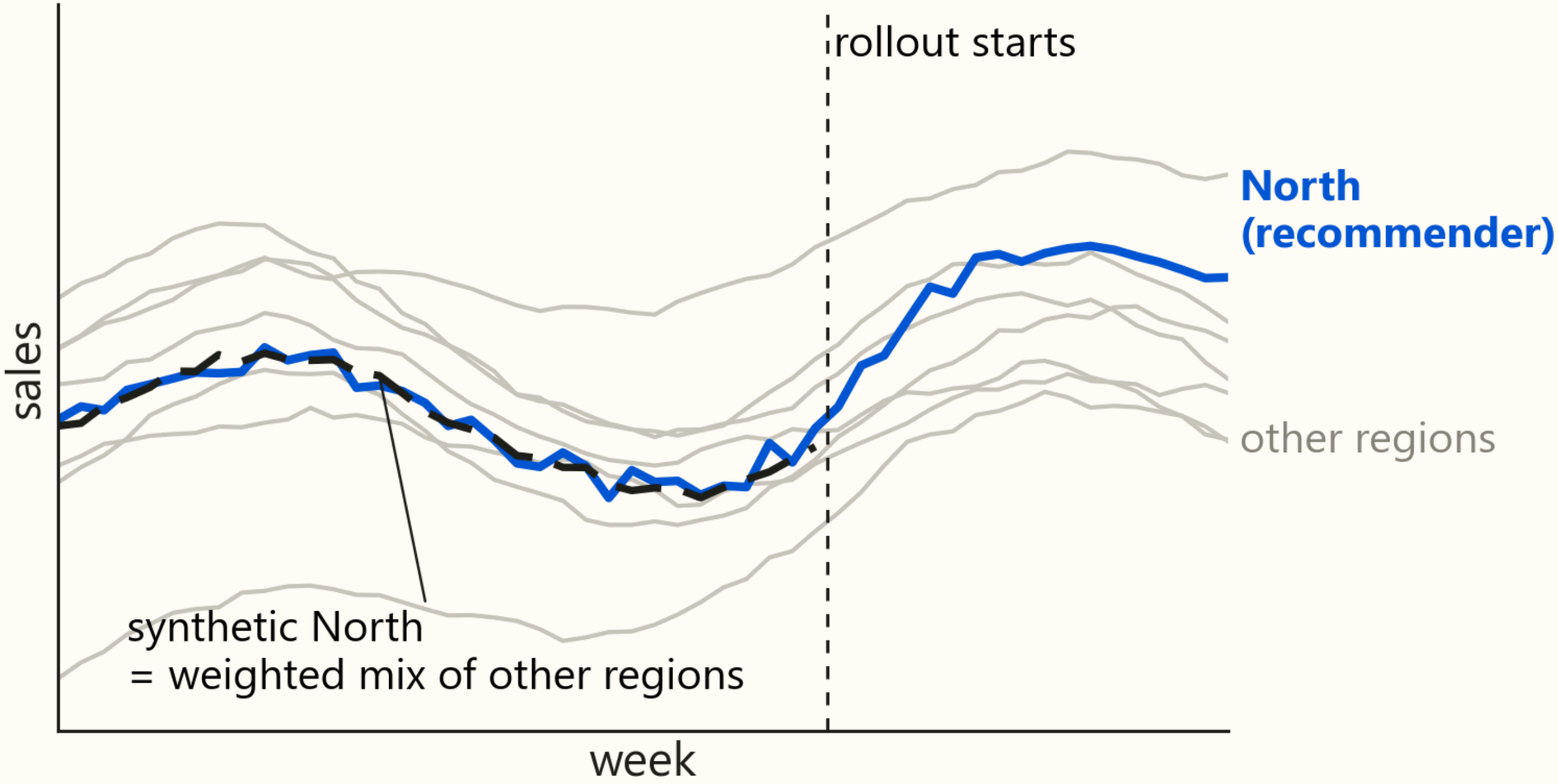
Let’s estimate the uplift with DML

Synthetic Control

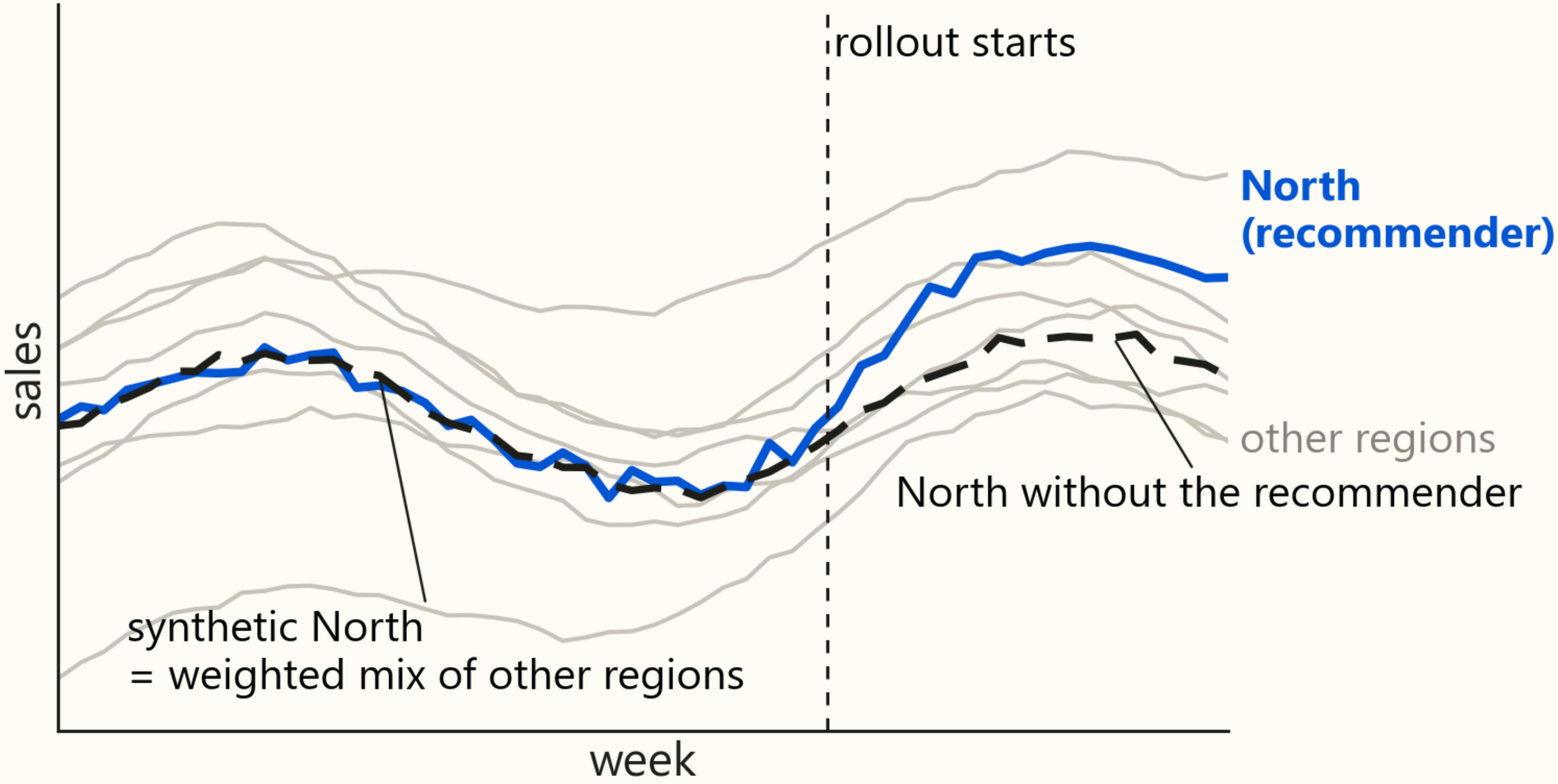
Synthetic Control



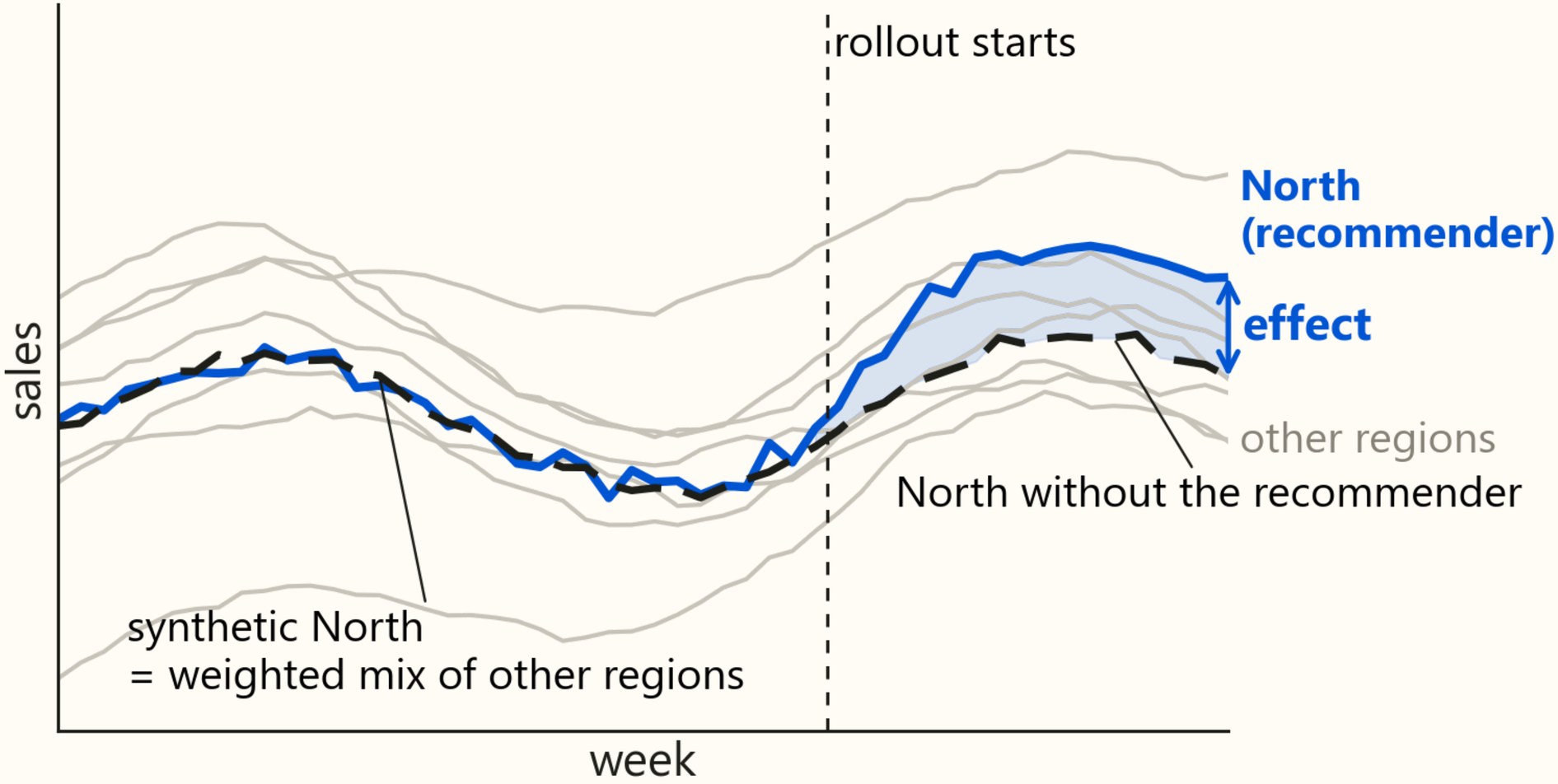
Synthetic Control



Synthetic Control



Synthetic Control



Synthetic Control

decision academy

Estimand: ATT

- Synthetic control estimates the **ATT**: the effect on the North, after the start
- ATE averages the effect over everyone, ATT only over those treated
- The two match when the effect doesn't differ between treated and untreated

$$\mathrm{ATT} = \mathbb{E}\left[\, Y(1) - Y(0) \mid T = 1 \,\right]$$

$$\widehat{\mathrm{ATT}}_{SC} = \frac{1}{T - T_0} \sum_{t > T_0} \left(Y_{\mathrm{North},\, t} - \sum_{j} w_j \, Y_{j,\, t} \right)$$

$$\mathrm{ATE} = p \cdot \mathrm{ATT} + (1 - p) \cdot \mathrm{ATC}$$

T_0: last period before the start · w_j: weight of donor region j · p: share of units treated · ATC: effect on the untreated

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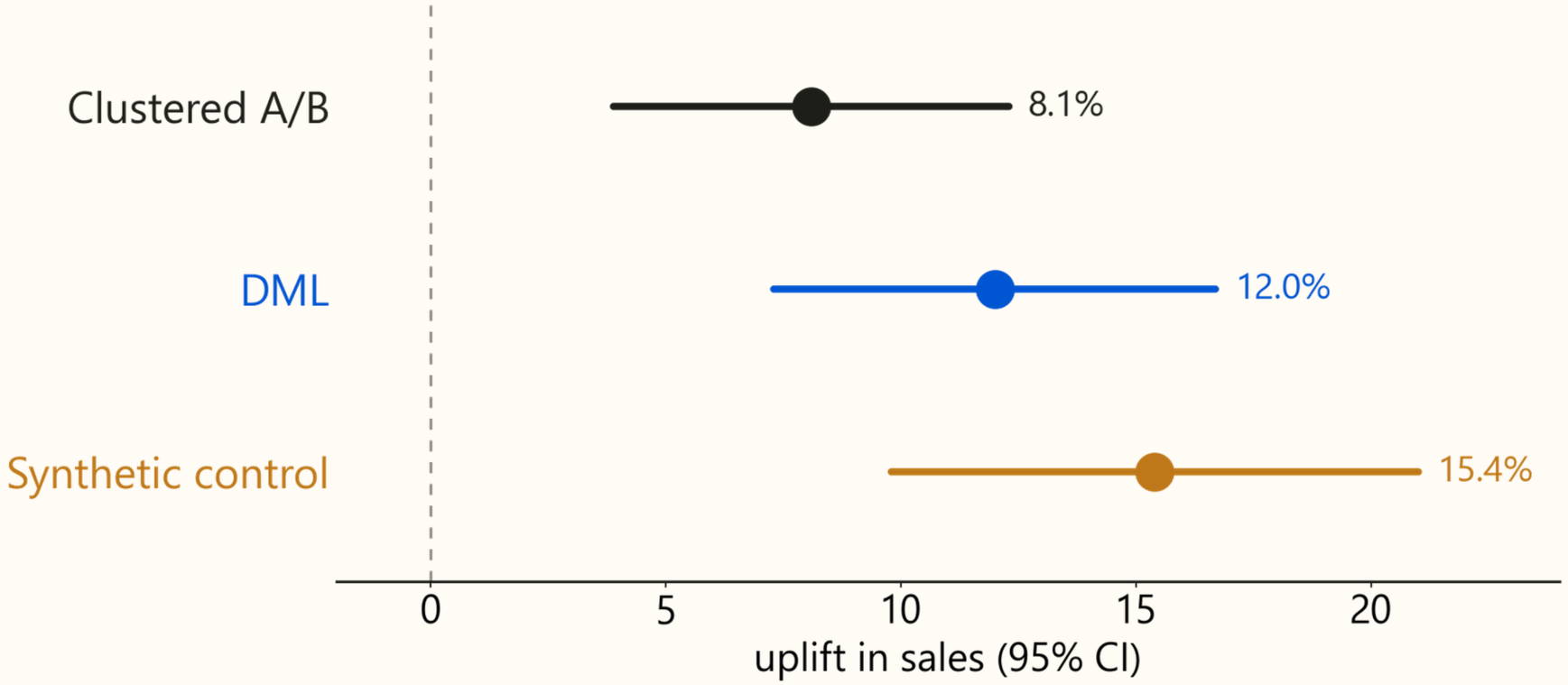
Let’s build a synthetic control

Triangulation

Triangulation

Why triangulate

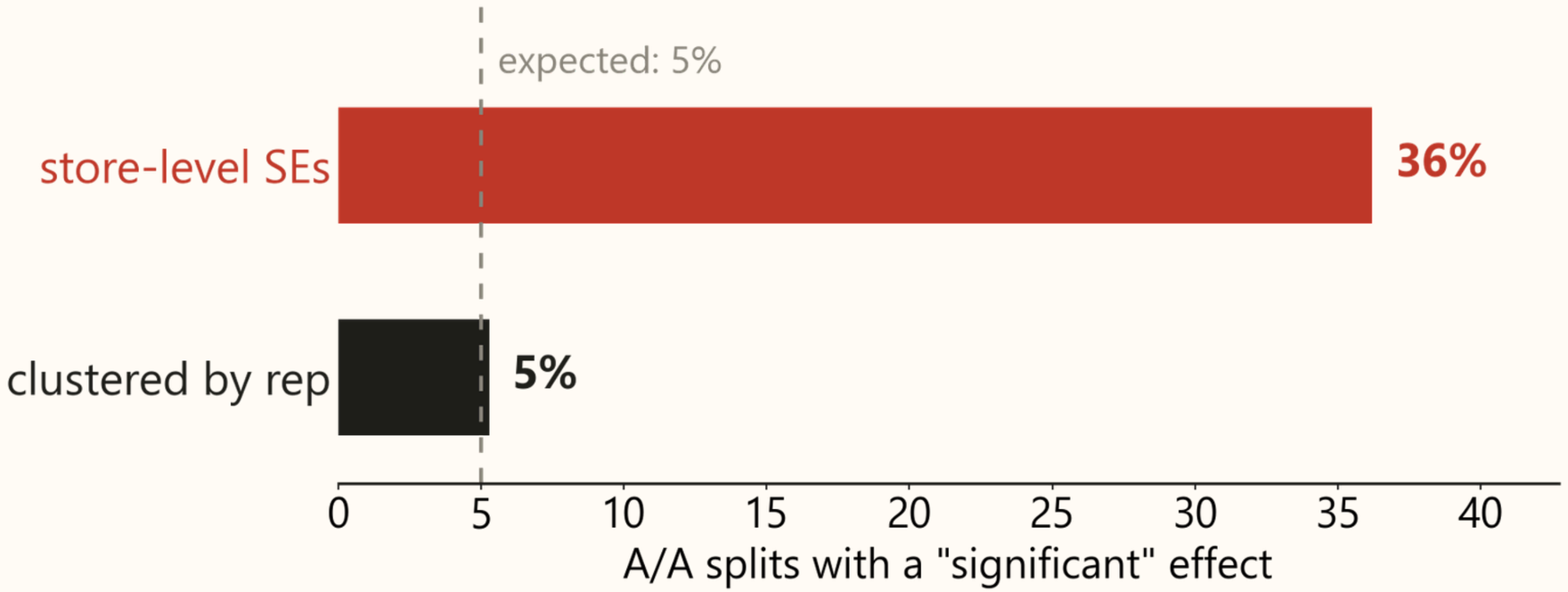
- Each method can fail, sometimes **in a different way**
- If methods with different assumptions agree, we will typically trust the result more



Triangulation

Stress test 1: A/A test

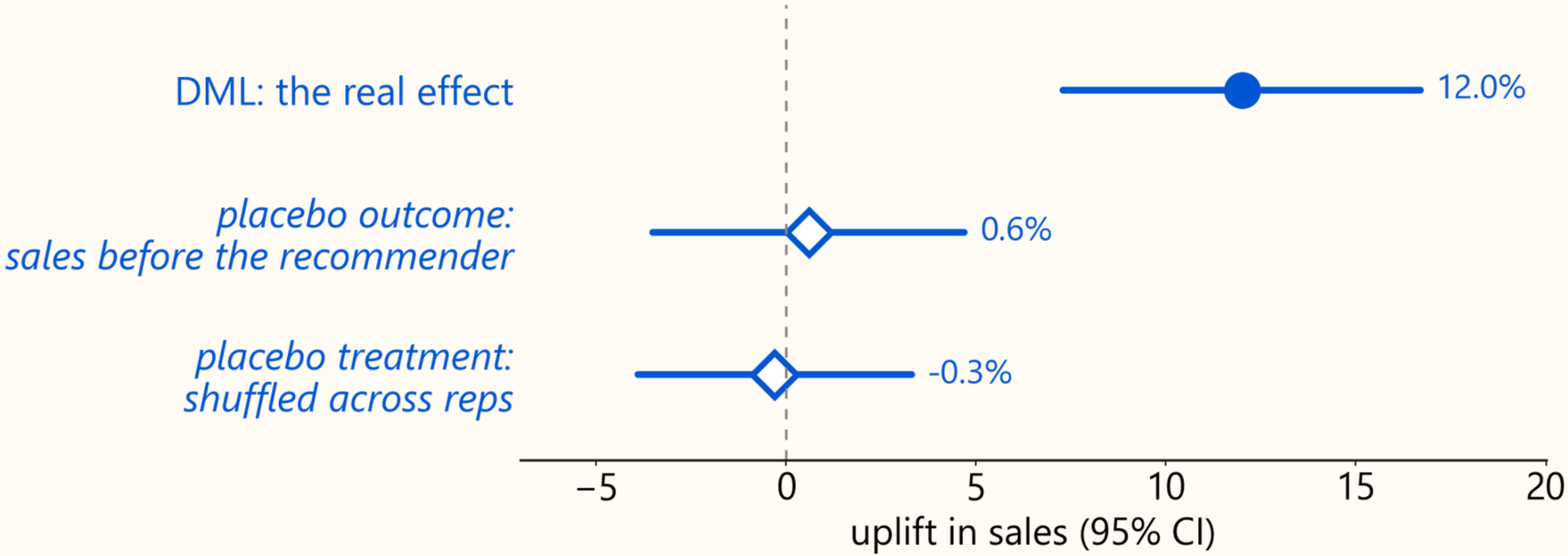
- Split the reps into two arms, **no treatment** in either
- Store-level SEs find an "effect" in 36% of splits. Clustered by rep: 5%



Triangulation

Stress test 2: DML placebos

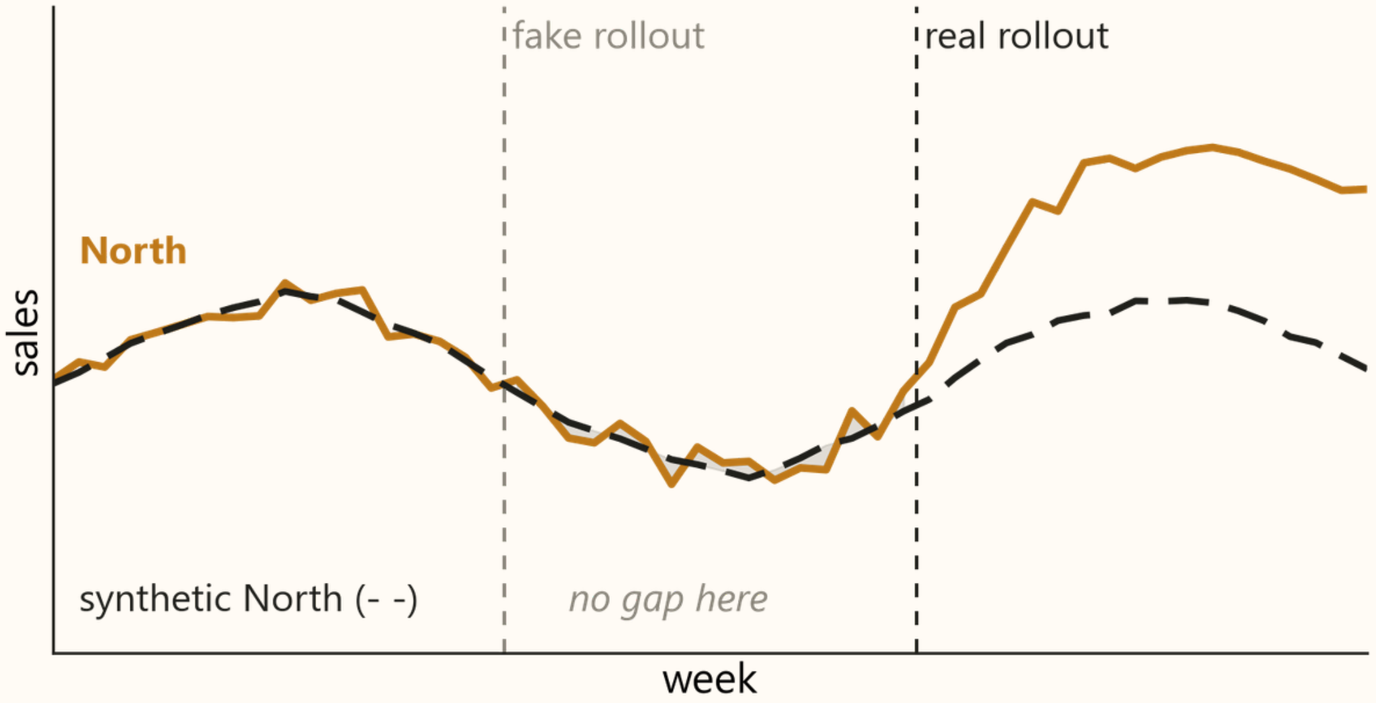
- Run DML where **the answer must be zero**
- A placebo outcome and a shuffled treatment should both land on zero



Triangulation

Stress test 3: SC with a fake rollout date

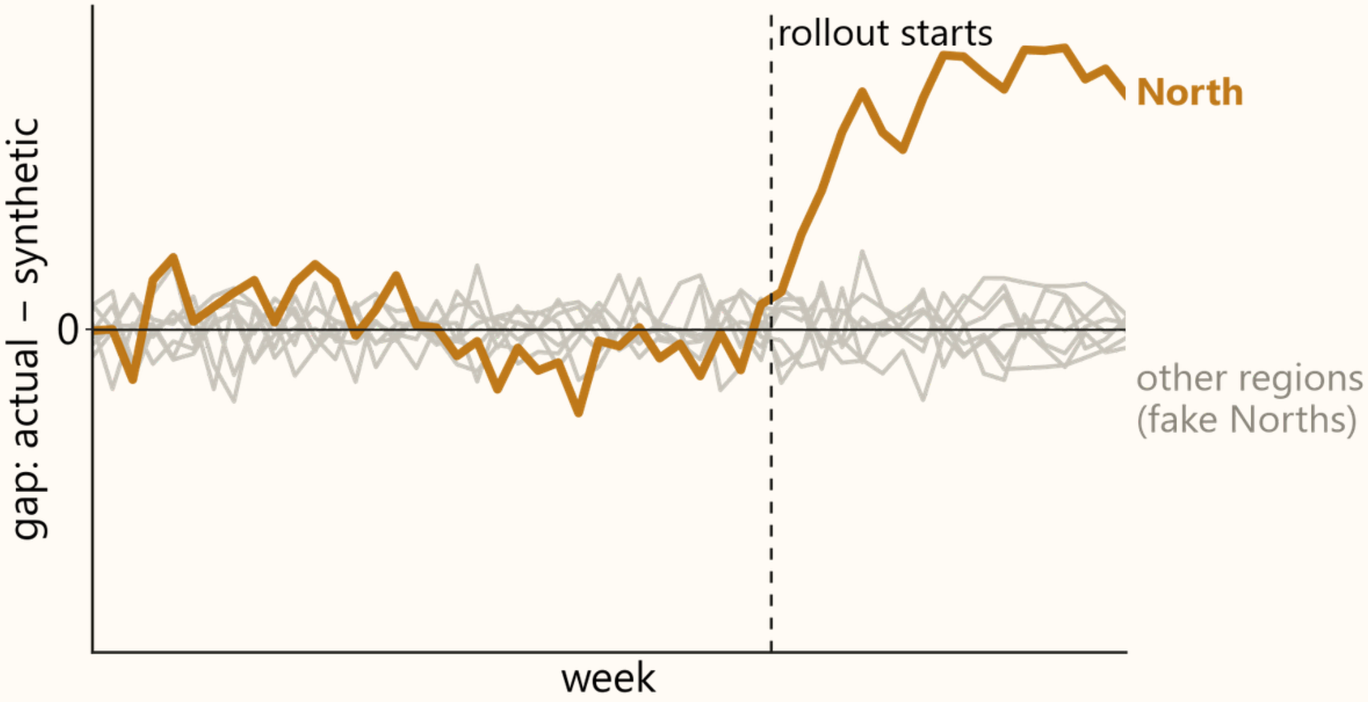
- Move the rollout date **into the pre-period**
- Fit only on the weeks before it: no gap should open until the real rollout



Triangulation

Stress test 4: SC with fake Norths

- Run SC for **every other region** as if it got the recommender
- The North's gap should stand out. With 8 regions, the best p-value we can get is 1/8



Triangulation

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Stress tests: the summary

- Each method got a case where **the true effect is zero**: did it say so?
- The one failure: store-level SEs, which ignore the randomization unit

	Test (true effect = 0)	Result	Pass?
Clustered A/B	A/A test, clustered by rep	5% false positives	✓
	A/A test, store-level SEs	36% false positives	✗
DML	placebo outcome	0.6% (−3.5 to 4.7)	✓
	shuffled treatment	−0.3% (−3.9 to 3.3)	✓
Synthetic control	fake rollout date	no gap before the real date	✓
	fake Norths (other regions)	North ranks 1 of 8 (p = 1/8)	✓

Triangulation

What we tell the business

- Give **a range**, not a single number
- Say which method we trust most, and why
- Name the weakest assumption
- Say what would settle it, e.g. a longer test where the estimates disagree

Triangulation

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Let’s test it!